

A PROJECT REPORT ON

**HUMAN-CENTRIC ROBOTIC CONTROL USING
SPEECH RECOGNITION AND VISION-BASED
OBJECT DETECTION**

**Submitted in partial fulfillment of requirements
for the award of the degree of**

**BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**

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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**

**RAJEEV GANDHI MEMORIAL COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)**

AFFILIATED TO J.N.T UNIVERSITY ANANTAPUR. ACCREDITED BY NBA (TIER-1) &
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RECOGNIZED UGC-DDU KAUSHAL KENDRA
NANDYAL-518501, (Estd-1995)

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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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CERTIFICATE

This is to certify that **BOMMASANI PEDDARAJU (22091A3338)**, **AKULA NITHESH (22091A3336)**, **KURUVA NIRMALA (22091A3335)** and of IV- B. Tech II- semester, have carried out the major project work entitled “**HUMAN-CENTRIC ROBOTIC CONTROL USING SPEECH RECOGNITION AND VISION-BASED OBJECT DETECTION**” under the supervision and guidance of **Dr. G. Kishor Kumar M.Tech, Ph.D**, Professor & HOD, Dept. of CSE (AI & ML), in partial fulfillment of the requirements for the award of Degree of **Bachelor of Technology** in **Computer Science and Engineering (Artificial Intelligence & Machine Learning)** from **Rajeev Gandhi Memorial College of Engineering & Technology (Autonomous)**, Nandyal is a bonafied record of the work done by them during 2025-2026.

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We hereby declare that the work done in this project entitled “**HUMAN-CENTRIC ROBOTIC CONTROL USING SPEECH RECOGNITION AND VISION-BASED OBJECT DETECTION**” submitted towards completion of major project in IV Year II Semester of B.Tech CSE (AI&ML) at the **Rajeev Gandhi Memorial College of Engineering & Technology, Nandyal** is an authentic record of our original work done under the esteemed guidance of **Dr. G. Kishor Kumar** M.Tech, Ph.D, Professor & HOD, Dept. of **COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**, RGM CET, Nandyal.

We have not submitted the matter embodied in this report for the award of any other Degree in any other institution for the academic year 2025-2026.

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Whatever one does, whatever one achieves, the first credit goes to the **Parents** be it not for their love and affection, nothing would have been responsible. We see in every good that happens to us their love and blessings.

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ABSTRACT

In the last few years, the combination of robotics and artificial intelligence has been a big step forward for automation and human-robot interaction in many fields[1]. Still, older robotic control systems are usually harder to use for people who aren't tech-savvy because they require more complicated programming and interfaces. To fix this problem, this project will suggest a human-centered robotic control system that uses speech recognition[2] and vision-based validation to make control more reliable and easy to understand.

The suggested system lets you control the robot arm with your voice[3]. A microphone captures the user's voice, and the speech recognition module converts the spoken input into text-based commands, which are then processed by a computer vision module is also added as a way to check that things are correct and avoid doing things by accident. A camera is placed above the workspace to watch it, and colored objects are found and detected using image processing techniques like HSV color segmentation[4], contour detection, and centroid calculation.

The system also uses homography-based coordinate transformation[5] to change image coordinates into real-world robotic workspace coordinates, which makes it possible to move objects with precision. The robotic arm will then follow the validated commands to pick up and move things with a suction gripper. This mix of speech processing, computer vision, and robotic control creates a multi-modal framework that makes the system more reliable, safe, and easy to use.

The experimental evaluation of the project demonstrates that the proposed system performs effectively and operates reliably in real-time environments. It can accurately understand voice commands, find things, and carry out robotic actions well in real-world situations. These results show that the system can work well in real life by combining speech recognition with vision-based detection.

The study also shows that the system makes it easier and more natural for people and robots to work together [1]. It also doesn't need a lot of processing power, which makes it easy and practical to use. Overall, the system that was created is smart and easy to use. It can be used in a variety of fields, including smart environments, industrial automation, educational robotics, and assistive systems.

Keywords: Computer Vision, Speech Recognition, HSV-Based Segmentation, Homography Transformation, Dobot Magician, Human–Robot Interaction, OpenCV, Pick-and-Place Operations, Voice-Controlled System

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
HRI	Human–Robot Interaction
CV	Computer Vision
HSV	Hue, Saturation, Value
RGB	Red, Green, Blue
API	Application Programming Interface
GUI	Graphical User Interface
ROS	Robot Operating System
CPU	Central Processing Unit
RAM	Random Access Memory
USB	Universal Serial Bus
NLP	Natural Language Processing
IoT	Internet of Things

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CHAPTER - 1

INTRODUCTION TO HUMAN-CENTRIC

ROBOTIC CONTROL SYSTEM

1.1 Overview of the System

In summary, the objective of this project is to develop a human-centered robots control system that facilitates a more natural human interaction with the robots. Robotics technologies, Artificial intelligence and computer vision have seen considerable developments within the last few years, and this has improved human interaction with machines [1]. Nevertheless, a majority of the conventional robots still use techniques such as keyboard, joysticks and programmed commands that are difficult for non-technical individuals to understand [2]. Also, they lack natural communication techniques hence are not as reliable during real time applications.

To tackle these challenges, human-centered robotic systems have been introduced to make interactions between humans and machines more intuitive. These systems allow users to communicate in natural ways, such as through speech and visual cues, improving overall usability [3]. By combining computer vision with speech recognition techniques, robots can accurately interpret user instructions and adjust their behavior according to changes in their surroundings [4].

This project introduces a system that utilizes speech recognition as the primary method for user input. Users commands are received in the form of audio and then converted into text with the help of Python library. In this way, users will be able to use easy and straightforward voice commands to manipulate the robotic system instead of relying on controllers and keyboards [5].

A vision-based module is employed to identify and locate objects within the working environment. In order to detect objects in real time, images from the camera feed are analyzed with OpenCV. The images are then transformed into HSV (Hue-Saturation-Value) colors, which makes object identification easy regardless of the lighting condition [6]. Objects of particular colors such as red, green and blue are detected with color segmentation techniques while contours are used to find the location of the object [7].

Homography mapping is employed in transforming coordinates of the image pixels into real life coordinates to enhance accuracy in the robotic movement [8]. In this scenario, the robotic arm can accurately position itself based on the detected object coordinates. The final task is carried out by a dobot magician robotic arm that is controlled by voice commands [9].

Integrating voice recognition with computer vision enhances the overall reliability and performance of the system. The robot will act on the commands after analyzing the validity of

the object and the commands. This will prevent mistakes and reduce risks while carrying out operations [10]. It should be noted that the system proposed is efficient and easy to use and can be modified for other applications.

By integrating the two techniques, the robotic system will be able to make better decisions taking into account both the user inputs and the environment [3]. This type of input is more efficient compared to the other single input system because it performs better in real life situations. The system will also be able to constantly monitor the surrounding while interacting with the robot [6]. Therefore, it provides a flexible and dynamic way of performing complex operations and manipulating modern day automation systems.

1.2 Characteristics of the System

There are several characteristics of this human-centered robotic control system that enhance its functionality, versatility and reliability in real time applications. One of the key features of this system is its ability to operate in real time. Real time processing of the input data, including speech and vision analysis allows this robotic system to act promptly to the commands given by the users. This is a highly important feature for any robot because even a slight delay may negatively affect performance. Consequently, low latency is a vital component of this system design [1].

Another key feature of the system is its multimodal capability, enabling it to process and combine multiple types of input, allowing it to process and integrate multiple types of input. Unlike other traditional robotic systems that depend on single input technique, this system combines the advantages of speech recognition with the use of computer vision to detect objects. Communication is simplified and becomes more natural because users can issue commands using the voice and the vision module determines whether there are available objects before executing. This combination will significantly reduce errors caused by wrong voice recognition, increasing overall system safety and reliability [3].

An additional key feature of the system is its ability to control the robot without any physical contact. By employing voice commands, the system allows users to operate the robot without having to touch any device. It is highly advantageous in places such as hospital, labs, and factories where safety and hygienic requirements are important. Using voice commands enables users to control the robotic arm without any danger [2].

Accuracy is another characteristic of the system. This system utilizes speech recognition that recognizes the voice commands with high accuracy in normal conditions. The vision system, on the other side, uses the HSV color segmentation and contours to detect objects based on their color. The use of HSV colors is more efficient compared to RGB as the system is robust to lighting changes [6].

Modular architecture of this system is an important aspect that distinguishes it from other systems. In this case, various modules, such as speech processing, vision processing, coordinate mapping, robotic control, etc. are included in the system that perform particular operations independently while communicating with each other. This way of designing, testing and maintaining is much easier than monolithic architectures, allowing future modifications [4].

Coordinate transformation is another unique capability incorporated into the system.. Through the use of homography, coordinates of the object in the pixel are mapped to real-world coordinates, which the robot will understand. As a result, the robotic arm will go exactly where it needs to pick or place the object [8].

In conclusion, the proposed system is developed with flexibility and scalability, allowing it to be easily extended in the future. More sophisticated algorithms can be incorporated to enhance object detection performance, while advanced voice recognition methods can be added to broaden its functionality. This adaptability ensures that the system can evolve and become more efficient over time [10].

In general, such characteristics of the system as real-time performance, multimodal operation, contactless control, high accuracy, modular architecture and adaptability enable its effective performance and flexibility in different applications. It is also important for human-robot contact, promoting robotic systems performance.

1.3 Applications of the System

The robotic control system based on voice recognition and vision analysis presented in this project can be employed in many practical applications. To begin with, this system can play a significant role in improving industrial automation processes. Robotic arms can be employed for picking and placing objects, sorting, handling materials, etc. With voice recognition, they will be easy to control and program, which will increase productivity in industrial settings [9].

Robotic system also plays an important role in the healthcare industry, where voice commands can be used to control robotic assistants and equipment contactlessly. In places where hygiene is of vital importance, it will be possible to perform operations without coming into physical contact with the equipment [2].

Another important application of this system is in the field of education, where voice-controlled robots can be effectively utilized. It enables students and researchers to gain hands-on experience in areas such as robotics, artificial intelligence, and computer vision. By bridging theoretical concepts with practical implementation, the system supports skill development and deeper understanding [1].

In addition, the system has potential applications in assistive technologies, particularly for individuals with physical disabilities. It allows users to control robotic operations through voice commands, such as picking and placing objects at desired locations, thereby helping them

perform everyday tasks more easily. This can significantly improve their independence and overall quality of life [3].

The system can also be employed in smart environments and home automation, helping to increase convenience. Voice commands will allow controlling multiple devices, while vision-based object recognition will improve system response rate. Thus, it will facilitate routine operations [4].

Moreover, robotic control system could be used in research and development, serving as a testing platform. The proposed robotic system provides a platform for experimenting with different approaches to human–robot interaction. Future system upgrades may also include such technologies as machine learning and deep learning [10].

Therefore, in general, this robotic control system can be successfully applied in many fields due to voice recognition and vision-based object detection capabilities.

1.4 Importance of the System

As discussed earlier, the significance of this human-centered robotic control system lies in its ability to enable more efficient and intuitive interaction between humans and robots. In general, robotic systems require complicated interfaces and programming skills, making it accessible only to technical experts. The system proposed enables users to control robotic devices with the help of speech recognition. This capability makes the system more user-friendly for individuals without technical expertise, thereby promoting wider adoption of robotic technologies [5].

Another important characteristic of the proposed robotic system is vision-based object recognition, which improves its reliability and safety. The system does not simply execute voice commands; however, it verifies the availability of an object in the vicinity by applying computer vision. This helps reduce the likelihood of errors while also enhancing safety during operations. Accuracy of the operation is highly important in many robotic applications [6].

The system also plays a significant role in improving the overall efficiency of work processes. Robotic systems will perform tasks faster compared to traditional methods, as the communication is based on voice. Faster performance rate and higher accuracy of operation make the system more effective in practical application, particularly in industrial settings [1].

Moreover, this robotic control system is malleable and adaptable, meaning that it can be adapted to different voice commands, kinds of objects and tasks. This allows its application in various industries, including healthcare, education, manufacturing, and assistive technologies [4].

The system's convenience and accessibility are also important factors to consider. In this regard, the system allows operating a robotic arm without having to come in physical contact with it. Such an ability will come in handy in hazardous or sanitary environments where direct contact is inappropriate. This characteristic also improves safety of the system [2].

Also, the system is characterized by the fact that it improves human-robot interaction by linking human desires to machine responses. It is more efficient because it uses both voice and vision analysis [3].

Overall, the system proposed has multiple advantages in terms of usability, accuracy, safety and improving human-robot interaction. This makes the system well-suited for contemporary robotics applications.

1.5 Advantages of the System

Several advantages can be outlined in connection with this robot control system that renders it beneficial in contemporary robotics. The possibility to interact naturally using voice commands is an important advantage. In comparison to traditional systems that require complex controllers and programming, this robotic system will be easier to use [3].

The ability to operate a robotic arm remotely without touching it is another important advantage. This characteristic will be especially valuable in environments, where it is hard or risky to come into contact with robotic devices. This aspect further enhances the safety of the operations carried out within the system [2].

The ability of the system to function in real time is another significant advantage. Fast processing of speech and visual input data allows for continuous and fluent operation of the system, contributing to improved performance rate [1].

Another key advantage of this robotic system is its relatively low production cost. There is no need for purchasing expensive hardware; instead, existing components, such as a camera and a microphone will be used. This reduces costs and makes the product affordable [4].

Accuracy of the operations is provided by combining speech recognition with vision-based object detection. Color segmentation and contour detection ensure accurate recognition of objects independent from lighting conditions. Finally, visual validation guarantees that only the right object is picked and placed [6].

The system is characterized by flexibility, which enables adding new functions, such as new methods of recognition and more voice commands, improving its performance and responsiveness [10].

Finally, the robotic control system proposed allows users to cooperate with the robot by issuing voice commands. This will enable them to perform actions more efficiently in workplace, where robots and humans are involved [9].

In conclusion, ease of use, safety, low cost, high accuracy and responsiveness, as well as flexibility, are some of the advantages of this robotic system.

1.6 Summary

In general, this chapter introduces an innovative human-centered robotic control system using speech recognition and vision-based object detection, allowing improving human-robot interaction. It enables users to control a robotic system by using voice commands eliminating the need for other conventional input devices. This is going to enhance the efficiency of such a system because it will become more accessible to various users, including non-technical [5].

The use of visual analysis further enhances the reliability and accuracy of the robotic system. Techniques such as HSV-based color segmentation and contour detection are applied to ensure accurate object identification. Additionally, the vision module verifies and validates the inputs before executing voice commands, improving overall system performance [6].

The chapter has also discussed the characteristics of the proposed system, such as real-time functioning, multimodal interaction and contactless control of robotic arms. These characteristics contribute to making the system suitable for practical application in various areas, such as industrial automation, healthcare, education and assistive technologies [1].

Among the important applications, this robotic system is also characterized by ease of use, low cost and high accuracy. It enables human-robot interaction through voice commands.

To conclude, the robotic system discussed is a valuable step in creating user-friendly and effective robotic devices [10].

CHAPTER – 2

LITERATURE REVIEW

2.1 Introduction to Literature Survey

The literature survey plays in this project to recognise the current research and technological advancements on human-centric robotic control systems. It gives a clear depiction of the methods proposed in the past and assists in their strengths and limitations. This knowledge can be applied in the development of a system that has the capability of addressing the current issues and deliver a better performance. Over the past few years, there has been a rapid growth in robotics, speech recognition and computer vision and it has resulted in more intelligent and user friendly robotic systems. By employing communication channels like voice recognition and visual perception as opposed to the conventional control interface, researchers have concentrated on the possibility of letting human beings interact naturally with machines [1].

Speech recognition has emerged as a key technology for enabling effective interaction between humans and robots. It enables users to interact with robotic systems in a natural language and hence the interaction is easy and user-friendly. The previous systems largely relied on cloud-based processing, which was highly accurate but created delays since it relied on the network. A way of overcoming these problems is by designing modern systems to function by using real time and offline processing methods. Such methods minimize delays and enhance responsiveness, and are therefore more applicable to real-world applications. The accuracy of the speech recognition systems is however, still impaired by challenges like the background noise, differences in accents and pronunciation differences. Thus, more validation techniques are needed to enhance reliability [5].

Computer vision techniques, along with speech recognition, are an important part of allowing robots to perceive the environment they are in. Vision-based systems involve the use of cameras and image processing algorithms to identify objects, extract their features and determine their locations. Color segmentation and contour detection are classic methods since they are easy, effective and can be used in real time. HSV color space enhances the accuracy in detection in various lighting conditions which makes the system more robust. Advanced techniques like deep learning are more accurate, but they are resource intensive and do not necessarily fit in a lightweight or real-time system [6].

The latest studies indicate that speech recognition and vision-based object detection are complementary to create more effective and reliable robots. In these multimodal systems, commands can be given by speech, and then the vision module will check the occurrence of the

objects and then perform the actions. This strategy minimizes mistakes and enhances safety as the commands are verified before the execution. The combination of various technologies also improves the functionality of the system and allows more complex functions in human-robot interaction [3].

The other significant point that is raised in the literature is the application of the coordinate transformation methods like homography mapping. The methods are employed to transform the position of objects in images into real-life coordinates that the robotic system can comprehend. Proper mapping is needed to have an accurate movement of the robotic arm and effective performance of tasks. System performance can be influenced by any fault of mapping and hence, it requires appropriate calibration and validation [8].

Besides this, the literature survey also highlights the necessity of having systems that are efficient, user friendly and flexible in various environments. Most of the systems that are in existence are either very precise but very complicated, or very simple and not as reliable. As such, there is a necessity to develop systems with a balance between accuracy, efficiency, and usability. Human-centric methods are based on making systems more accessible by enabling users to communicate with the system through natural inputs like speech and visual feedbacks.

In general, the literature review is informative about the progress of speech-based, vision-based, and hybrid robots. It aids in finding the main issues and directs the creation of a system that is more reliable, efficient, and easy to use. The results of the survey are the basis of the proposed work that seeks to enhance human-robot interaction by employing a speech recognition system along with vision-based object detection.

2.2 Related Works

The associated work study in the project will be geared towards understanding how the human-robot interaction systems have evolved over time. The earlier systems of the robots were mostly operated manually and had predetermined commands, therefore, could not be easily manipulated by the non technical operators. With the advancement in technology, researchers devised some speech-based control mechanisms so that they could make interaction more natural and easy to use. These systems improved accessibility and reduced the level of complexity in operations by facilitating human interaction with robots through voice commands [5].

Initially, speech recognition systems were cloud-based applications, where on remote servers, audio data was processed. They were highly precise systems but had delays because they were network reliant and were not applicable to real time applications. The modern systems resolve these issues by means of local or offline processing which reduces the latency and also improves response time. However, such problems as noise, speech dissimilarities, and articulation discrepancy still affect the functioning of the systems.

In the meantime, systems of object detection based on visions have also significantly improved. Traditional image processing methods are color segmentation and contour detection which are feasible and simple. The HSV color segmentation is particularly useful since it can be done under varying lighting conditions. State-of-the-art deep learning technologies are more precise but require extensive computing resources and cannot be used in real-time operations [6].

The new researches state the necessity to implement a combination of different technologies. Multimodal systems are speech recognition systems that are integrated with vision detection to make them more reliable. It has these systems where it interprets the commands through speech and verifies the existence of the objects through the vision before it can be carried out. This will reduce the errors and maximize safety, thereby improving efficiency of the system in a real life scenario [3].

Another important feature that ensures successful mapping of image coordinates to real-world locations is coordinate mapping. Homography mapping techniques are of utmost use towards this aim since they provide precise transformation between the robot workspace and the camera view [8].

Generally, integrated robots systems are robots that integrate speech, vision and control into a single platform. The system finds applications in various areas such as automation, education, and research which is supported by flexibility, scalability and usability through this integration [10].

2.2.1 Speech-Based Robotic Control Systems

Speech-based robotic control systems are taken as a significant solution to enhancing a human-robot interaction in this project. These robots enable them to manipulate robots with easy voice commands, rather than conventional input devices like keyboards, joysticks or pre-written programs. This approach makes the system easier to use and accessible to individuals without technical expertise [3].

Previously, speech controlled robotic systems largely relied on cloud based speech recognition services. In this kind of systems audio input was taken via a microphone and sent to remote servers where it was decoded and turned into text. These systems, though had good accuracy because they had strong processing capabilities, were limited in a number of ways. They needed a steady internet connection, added latency since data was being transferred and could not be used in real time applications where timely response was required [5].

Modern systems are overcoming these limitations by using local or offline methods of speech recognition. Such systems can take voice input directly on the device without the need to have internet connectivity. Consequently, they minimize latency and enhance response time, which makes them more appropriate in controlling robots in real-time. This has enhanced the rapid response of the robot to user commands, which is essential in automation and handling of objects.

However, speech recognition systems still face challenges when applied in real-world environments. The command recognition accuracy may be influenced by factors like background noise, accent variations, speaking speed variations, and poor pronunciation. To solve these problems, noise filtering, confirmation of commands and error handling techniques are employed to enhance system reliability [2].

Moreover, newer advancements in natural language processing have improved the ability of the speech-based systems. Such systems have now been able to comprehend the differences in commands and offer a more flexible interaction. This enables users to issue commands in various forms and yet get the desired outcome and hence making the system more convenient and efficient [1].

The other significant advancement is that there is an incorporation of speech recognition in conjunction with other technologies like computer vision. In such systems, voice commands are verified using visual data before being executed. This helps in making sure that only the right conditions are followed when performing actions to reduce the number of errors and enhance safety [4].

In general, speech-based robotic control systems offer a more natural, efficient and convenient means of communicating with robots. Although there are some difficulties, these systems are becoming more reliable and can be implemented into the real world because of constant advances in speech processing and integration with other technologies.

2.2.2 Vision-Based Object Detection Systems.

Object detection systems based on vision are important in this project as they will ensure that the robots learn to perceive and communicate with their environment. Vision-based systems, unlike traditional systems, rely on cameras and image processing technology to capture, recognize and track objects in real time, rather than relying on predefined instructions. This will enable robots to be smarter in undertaking tasks and adapting to various environments, thereby becoming more useful in real-world scenarios [6].

Most vision-based systems in earlier stages were based on a classical image processing like color segmentation, thresholding, edge detection and contour analysis. One commonly used approach is color-based detection using the HSV (Hue, Saturation, Value) color space, which is preferred because it is less sensitive to variations in lighting conditions. This ensures that object detection to various lighting conditions is more accurate than when using RGB based means. Once the image is segmented, the boundaries of the objects are identified using contour detection methods and the position of the objects in the frame is calculated [7].

These conventional approaches are easy, effective, and applicable to real-time applications, particularly where they are controlled like in a laboratory or an educational system. But lighting change, shadows, background noise, and occluding an object can influence their performance.

Such restrictions have the potential to decrease the accuracy of detection in complex or dynamic environments.

To improve system performance, advanced approaches based on artificial intelligence and deep learning have been introduced. Detection of objects can be more precise and complex scenarios can be approached with the help of methods like convolutional neural networks (CNNs). Those methods are however more demanding in terms of large datasets, high computing power and processing time, thus, less applicable in lightweight or real time systems that need speed [6].

In most practical scenarios, less complex algorithms such as the HSV color segmentation are still used as they offer an acceptable tradeoff between accuracy and performance. These techniques are simple to apply, use fewer computational resources and could work well in real-time scenarios provided the environment is relatively controlled.

The other notable attribute of the vision based systems is their capacity to give spatial information of objects. The system identifies the position of objects within the image frame by computing their centroid, which helps in accurately determining their location. The information is vital in the control of the robotic arm to interact with the objects in a precise manner. The system, coupled with coordinate transformation methods like homography mapping, can be used to map image coordinates to real-world locations to allow robots to move with precision [8].

In general, vision-based object detection is essential for enabling advanced capabilities in modern robotic systems. They enable the robots to sense their surroundings, confirm user instructions and execute the duties with more accuracy and reliability. Despite the existing challenges, there are on-going advancements in computer vision methods which are increasing their capabilities and application in areas like robotics, automation, healthcare, and smart systems [10].

2.2.3 Multimodal Human–Robot Interaction Systems

In this project, multimodal human–robot interaction systems are used to improve the overall performance and reliability of robotic control. These systems combine multiple input methods such as speech recognition and computer vision, instead of depending on a single mode of interaction. By integrating different technologies, the system can better understand user commands and respond more accurately in real-world situations [3].

In a typical multimodal system, speech recognition is used to capture user commands in a natural and simple way. Users can give instructions using voice, which makes the system easy to use and accessible. However, relying only on speech input can sometimes lead to errors due to factors such as background noise, unclear pronunciation, or differences in accents. To overcome this limitation, a vision module is added to verify the command before execution.

The vision system continuously monitors the environment using a camera and checks whether the required object is present. For example, if a user gives a command to pick a specific object, the

system first detects the object using vision-based techniques before performing the action. This validation process ensures that the robot performs the correct task and avoids incorrect or unsafe operations [4].

One of the major advantages of multimodal systems is their ability to improve system reliability. If one input method fails or produces uncertain results, the system can rely on another input source. This makes the system more robust and suitable for real-world environments where conditions are not always controlled [2].

Another important benefit is improved flexibility. Users can communicate with the system more naturally, without the need to follow rigid command structures. By integrating inputs from both speech and vision, the system is able to make more accurate and informed decisions. This allows more complex tasks to be performed efficiently, such as selecting objects based on color or position.

Multimodal systems also enhance safety by reducing the chances of incorrect actions. Since commands are verified before execution, the system ensures that robotic movements are accurate and appropriate for the given situation. This is especially important in applications where precision and safety are critical, such as industrial automation and assistive robotics.

These systems are widely used in different fields, including smart environments, healthcare, industrial automation, and educational robotics. In collaborative environments, multimodal interaction helps humans and robots work together more effectively by improving communication and coordination [9].

Overall, multimodal human–robot interaction systems provide a more advanced and reliable approach compared to single-input systems. By combining speech recognition and vision-based detection, the system becomes more intelligent, accurate, and user-friendly. This makes it highly suitable for modern robotic applications and future developments in intelligent automation [10].

2.2.4 Coordinate Mapping Techniques

In this project, coordinate mapping techniques play an important role in enabling accurate interaction between the vision system and the robotic arm. When objects are detected using a camera, their positions are obtained in image coordinates (pixel values). However, these coordinates cannot be directly used by the robotic system. Therefore, it is necessary to convert image coordinates into real-world coordinates so that the robot can accurately reach and manipulate the object. This transformation process is known as coordinate mapping [8].

A widely adopted approach for this task is homography mapping, which defines the relationship between the camera's image plane and the robot's working area. During calibration, corresponding points from both the captured image and the actual workspace are identified. Using these point pairs, a transformation matrix is computed. This matrix enables the conversion of image pixel

coordinates into real-world coordinates, allowing the robotic arm to accurately interpret positions and perform movements accordingly [8].

Accurate calibration is very important in coordinate mapping. If the selected points are not precise or if there is any error in the transformation matrix, the robot may move to an incorrect position. Even small errors can affect the accuracy of pick-and-place operations. Therefore, careful calibration and testing are required to ensure that the mapping process is reliable and precise [9].

Another important factor in coordinate mapping is handling variations in the environment. In real-world applications, the position of objects, camera alignment, and workspace setup may change. The system should be able to adapt to these variations to maintain accuracy. In some cases, additional correction techniques or recalibration may be required to ensure consistent performance [7].

Coordinate mapping also works closely with other system components such as object detection and robotic control. Once the vision system detects an object and calculates its position in the image, the mapping technique converts this position into real-world coordinates. These coordinates are then sent to the robot control module, which generates the required motion commands for the robotic arm. This integration ensures smooth communication between different parts of the system and enables accurate execution of tasks [4].

Overall, coordinate mapping techniques such as homography are essential for bridging the gap between visual perception and physical action in robotic systems. They ensure that objects detected by the camera can be accurately reached by the robotic arm, improving precision and reliability. As robotic systems continue to develop, more advanced mapping techniques are expected to further enhance accuracy and adaptability in complex environments [10].

2.2.5 Integrated Robotic Control Systems

In this project, integrated robotic control systems represent a complete approach where multiple technologies such as speech recognition, Computer vision and robotic manipulation are integrated to form a unified system. Unlike traditional robotic systems that rely on only one method of control, integrated systems use different modules that work together to perform tasks efficiently. This combination improves system performance and allows better interaction between humans and robots [10].

In such systems, speech recognition is used as the primary method for receiving user commands. The system captures voice input through a microphone and converts it into text, which is then interpreted to determine the required action. This makes interaction simple and natural, as users can control the robot using voice instead of complex interfaces [3].

At the same time, the vision module plays a crucial role in understanding the surrounding environment. It utilizes a camera to identify objects and determine their positions accurately.

Before carrying out any operation, the system cross-checks the given command using visual data. This ensures that the robot proceeds only when the intended object is detected, thereby improving both accuracy and operational safety [4].

The robotic control module is responsible for performing the final action. Once the command is validated, it generates the necessary movement instructions for the robotic arm to complete the task. These instructions guide the robot to perform tasks such as picking and placing objects. Coordinate mapping techniques are used to ensure that the robot moves accurately to the desired location [8].

Another important feature of integrated systems is their modular design. Each component, such as speech processing, vision detection, and robotic control, operates independently but communicates with other modules. This simplifies the processes of designing, testing, and maintaining the system. It also allows new features to be added without affecting the entire system, improving flexibility and scalability [4].

Integrated robotic systems are widely used in various applications, including industrial automation, educational systems, research laboratories, and assistive technologies. In industries, they help automate repetitive tasks, while in education, they provide a platform for learning robotics and artificial intelligence. Their ability to combine multiple technologies makes them suitable for a wide range of applications [9].

Overall, integrated robotic control systems provide an efficient and intelligent solution for modern robotic applications. By combining speech recognition, vision-based detection, and robotic control, they improve system accuracy, reliability, and usability. This integrated approach plays a key role in advancing human–robot interaction and supports the development of next-generation robotic systems [10].

2.3 Challenges

In this project, multiple challenges arise while developing a dependable and efficient human-centered robotic control system. One of the key challenges is ensuring the accuracy of speech recognition. In practical environments, elements such as background noise, variations in accents, differences in speaking pace, and unclear pronunciation can reduce the system's ability to correctly interpret voice commands. Misinterpretation of commands may result in unintended robotic actions. Hence, enhancing the robustness of speech recognition under diverse conditions is a critical requirement for the system [5].

Another significant challenge is related to vision-based object detection. While methods like HSV color segmentation and contour detection perform well in controlled settings, their effectiveness can decline under real-world conditions. Factors such as varying lighting, shadows, reflections, and complex backgrounds can influence how objects appear, making accurate detection more difficult. Additionally, situations like object occlusion, where one object partially hides another,

can further reduce detection accuracy. These issues highlight the need for more robust vision systems that can operate reliably in diverse environments [6].

Latency is also a critical issue in real-time robotic systems. The system consists of several stages, including speech processing, image analysis, command interpretation, coordinate transformation, and robotic control. If any of these processes take too much time, it can delay the system response and reduce efficiency. Therefore, optimizing processing time and ensuring fast communication between modules is essential to maintain real-time performance [7].

Another important challenge is coordinate mapping accuracy. The system must correctly convert image coordinates into real-world coordinates so that the robotic arm can reach the exact position of the object. Any error in calibration or transformation can lead to incorrect positioning, which affects the accuracy of pick-and-place operations. Even small errors can have a significant impact on system performance, especially in tasks that require precision. Proper calibration and regular validation are necessary to overcome this issue [8].

System integration is also a complex challenge. The proposed system consists of multiple modules such as speech recognition, vision processing, and robotic control, which must work together smoothly. Ensuring proper communication and synchronization between these modules is difficult, especially when dealing with real-time data. A failure in one module can affect the entire system, making integration a critical aspect of system design [4].

Furthermore, enabling the system to function effectively in dynamic environments presents an additional challenge. Real-world conditions are not always predictable, and the system must be able to handle changes such as movement of objects, variations in lighting, and user behavior. This requires the system to be flexible and capable of handling different situations without affecting performance.

Furthermore, hardware limitations can also impact system efficiency. Limited processing power, memory constraints, and camera quality can affect both speech recognition and vision-based detection. Therefore, selecting appropriate hardware and optimizing software are important for achieving better performance.

Overall, these challenges highlight the need for continuous improvement in speech processing, computer vision, system integration, and real-time performance. Addressing these issues is essential for developing a reliable, accurate, and efficient human-centric robotic control system that can operate effectively in real-world applications.

2.4 Summary

In this chapter, a detailed review of existing research related to human-centric robotic control systems is presented. The literature survey focuses on key areas such as speech-based robotic control, vision-based object detection, multimodal interaction systems, coordinate mapping

techniques, and integrated robotic control systems. These studies show how robotic systems have evolved from traditional manual control methods to more advanced approaches that support natural interaction using speech and visual inputs [1].

The review of speech-based systems highlights how voice commands make robotic control easier and more accessible. Earlier cloud-based systems provided high accuracy but introduced delays due to network dependency. Modern approaches use real-time and offline processing methods, which improve response time and make the system more suitable for practical applications. However, challenges such as noise and variations in speech still need to be addressed [5].

Similarly, vision-based object detection systems were analyzed, focusing on techniques such as HSV color segmentation and contour detection. These methods are efficient and suitable for real-time applications, although their performance can be affected by environmental conditions such as lighting and background complexity. The use of vision systems improves the robot's ability to understand its surroundings and perform tasks more accurately [6].

The chapter also discussed multimodal interaction systems, which combine speech and vision to improve system reliability. By verifying commands using visual input, these systems reduce errors and enhance safety. In addition, coordinate mapping techniques such as homography mapping are essential for converting image coordinates into real-world positions, enabling accurate robotic movement [8].

Integrated robotic control systems bring together all these technologies into a single framework. This integration improves system performance, flexibility, and usability, making it suitable for various applications such as industrial automation, education, and research [10].

The challenges discussed in this chapter, including speech recognition accuracy, vision-based detection limitations, latency, and system integration issues, highlight the need for continuous improvement. Based on the insights gained from this literature review, the proposed system is designed to combine speech recognition and vision-based object detection to create a reliable, efficient, and user-friendly robotic control system suitable for real-time applications.

CHAPTER 3

SYSTEM DESIGN

3.1 Problem Definition

In this project, the primary issue addressed is the complexity involved in controlling robotic systems through conventional methods. Most existing systems rely on input devices such as keyboards, joysticks, or pre-defined programming, which often require technical expertise and are not easily accessible to non-technical users. As a result, interaction between humans and robots becomes complicated and less efficient.

Another limitation of traditional approaches is the absence of natural communication. Humans typically prefer intuitive interaction methods like speech and visual perception; however, many robotic systems do not support these modes. This limitation makes real-time control more difficult, especially in situations where quick and simple interaction is necessary.

Furthermore, systems that depend on a single mode of input, such as only speech or only vision, may not perform consistently under all conditions. Speech-based systems can be affected by background noise, while vision-based systems may struggle with variations in lighting or complex surroundings. These factors can negatively impact the overall reliability and accuracy of robotic operations.

Another challenge is achieving precise object detection along with accurate robotic movement. Without proper coordination between sensing and control mechanisms, the robot may either fail to identify the correct object or may not reach the intended position during tasks such as pick-and-place operations.

Additionally, many existing systems lack proper validation before executing actions. This can result in incorrect or potentially unsafe movements, particularly in dynamic environments. Therefore, ensuring higher levels of safety and reliability remains a critical requirement for such systems.

To address these problems, there is a need for a system that allows natural interaction, combines multiple input methods, and ensures accurate and reliable operation. The proposed solution focuses on developing a human-centric robotic control system that integrates speech recognition with vision-based object detection. This approach aims to simplify user interaction, improve system accuracy, and enhance overall performance in real-time applications.

3.2 Proposed Methodology

In this project, the proposed methodology focuses on developing a human-centric robotic control system by combining speech recognition with vision-based object detection. The aim is to create a system that allows users to interact with the robot in a simple and natural way while ensuring accurate and reliable task execution. By integrating multiple technologies, the system improves usability and overall performance compared to traditional robotic control methods [1].

The process starts by capturing user input through speech. A microphone records the user's voice commands, which are then processed using speech recognition techniques. The audio signal is converted into text with the help of Python-based libraries, and the system interprets this text to identify the intended action. This method allows users to interact with the robot using natural language, eliminating the need for complex input devices and improving ease of use [5].

Once the voice command is interpreted, the vision module is activated to analyze the workspace. A camera captures real-time images, which are processed using OpenCV. These images are transformed into the HSV color space to achieve better performance under different lighting conditions. Through color segmentation, objects such as red, green, and blue are detected. Techniques like contour detection and centroid calculation are then applied to accurately determine the position of these objects within the scene [6][7].

After identifying the object, coordinate transformation is carried out to convert image-based coordinates into real-world coordinates. This is accomplished using homography mapping, which establishes a relationship between the camera's perspective and the robot's working area. This transformation ensures that the robotic arm can precisely reach the object's location, enhancing operational accuracy [8].

Before performing any action, the system includes a validation stage. Both the interpreted voice command and the detected object are cross-checked to confirm their correctness. This step minimizes the risk of incorrect or unsafe operations, thereby increasing the overall reliability of the system. The command is executed only after successful validation [3].

Finally, the robotic control module generates the required movement instructions for the robotic arm. The robot moves toward the detected object, grasps it using a suction gripper, and places it at a predefined location. The entire sequence is executed in real time, ensuring smooth, accurate, and efficient task completion [9].

The system is designed using a modular architecture, where each component such as speech processing, vision detection, coordinate mapping, and robotic control operates independently but communicates with other modules. This modular design simplifies development, testing, and maintenance, and also allows easy integration of future improvements [4].

Overall, the proposed methodology combines natural human interaction with intelligent perception and precise robotic control. This integrated approach improves system accuracy, efficiency, and usability, making it suitable for real-world applications such as industrial automation, educational systems, and assistive robotics [10].

3.2.1 System Architecture

The architecture of the proposed human-centered robotic control system is designed to combine speech recognition, vision-based object detection, and robotic control within a unified framework.

It is composed of multiple interconnected modules, each responsible for a specific task, ensuring efficient operation and smooth data flow in real time [4].

The first module is the speech recognition unit, which captures user input through a microphone. The spoken commands are processed and converted into text using speech recognition techniques. This text is then analyzed to determine the intended action, such as identifying a particular object. This module enables intuitive and natural communication between the user and the robotic system [5].

The following module is dedicated to vision-based object detection. A camera captures real-time images of the workspace, which are processed using OpenCV. These images are transformed into the HSV color space to enhance detection performance under varying lighting conditions. The system then applies color segmentation and contour detection methods to identify objects and determine their positions within the frame [6][7].

Once the object is detected, the coordinate mapping module converts image coordinates into real-world coordinates through homography mapping. This transformation establishes a relationship between the camera view and the robot's workspace, allowing the robotic arm to accurately locate and reach the object. Proper calibration is carried out to maintain precision in positioning [8].

The command interpretation module acts as a bridge between input and execution. It matches the recognized voice command with the detected object to ensure that the correct task is performed. This validation step confirms both the command and object detection before execution, thereby improving system reliability and reducing the possibility of errors [3].

Finally, the robotic control module performs the required action. It receives processed information and generates movement instructions for the robotic arm. The robot moves to the target location, grasps the object using a suction mechanism, and places it at a predefined position. All operations are carried out in real time, ensuring efficient and smooth task execution [9].

All modules are organized in a structured manner to form a complete system. The modular design allows each component to function independently while maintaining coordination with others. This approach makes the system flexible, scalable, and easier to maintain, while also allowing future enhancements without affecting overall performance [10].

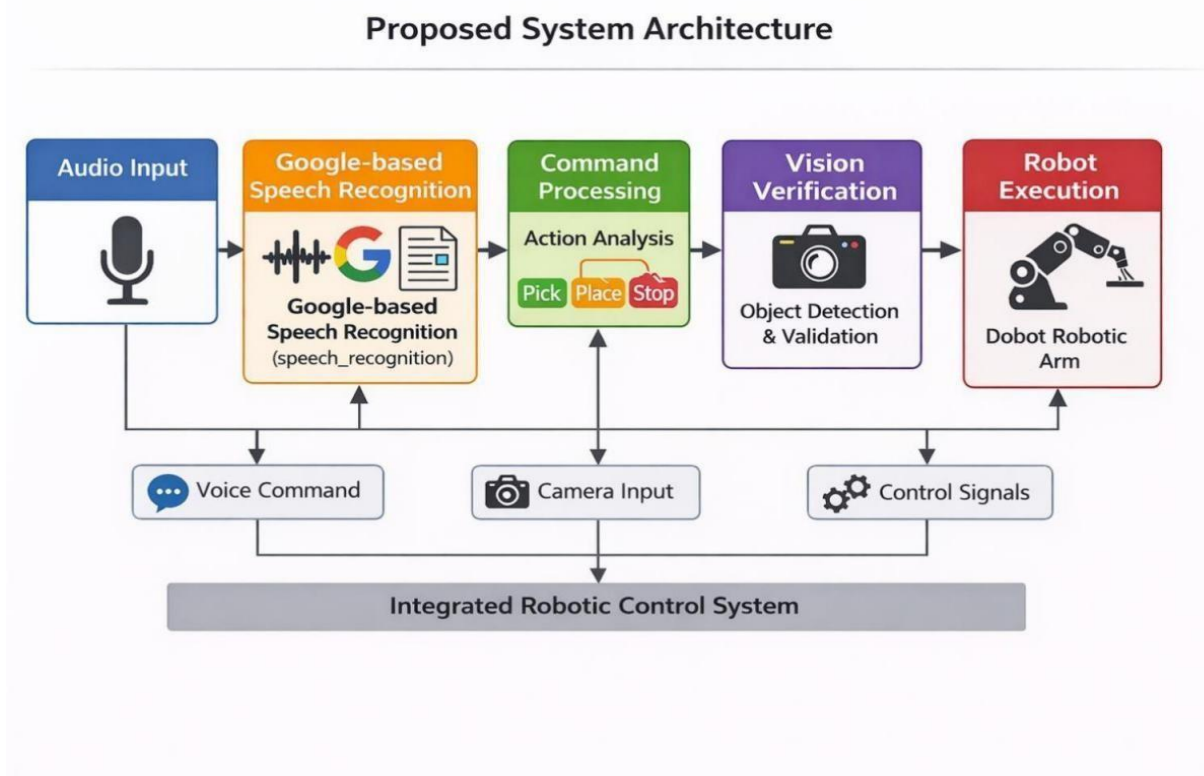


Fig 3.2.1: System Architecture

3.3 Modules

In this project, the overall system is organized into multiple modules, each designed to perform a specific task for smooth and efficient operation. This modular structure simplifies development, testing, and maintenance, while also allowing new features to be added without affecting the existing system. The key modules include speech recognition, vision-based object detection, command interpretation, and robotic control [4].

The speech recognition module processes user input received through voice commands. A microphone captures spoken commands, which are then processed using speech recognition techniques. The audio signal is converted into text using Python-based libraries, and the system analyzes this text to determine the intended action. This module enables a natural and user-friendly way for humans to interact with the robot [5].

The vision-based object detection module is responsible for identifying objects within the workspace. A camera captures real-time images, which are processed using OpenCV. These images are converted into the HSV color space to achieve better detection under different lighting conditions. By applying color segmentation and contour detection methods, the system identifies objects and calculates their positions accurately [6][7].

The command interpretation module acts as an interface between input and execution. It matches the recognized speech command with the detected object to ensure the correct selection. Additionally, it performs a validation step to confirm that both the command and detection results are accurate before proceeding. This reduces the chances of errors and enhances system reliability [3].

The robotic control module executes the final operation of the system. It receives the processed information, including the object's position and the corresponding command, and generates movement instructions for the robotic arm. The robot then moves to the target location, picks the object using a suction gripper, and places it at a predefined position. This ensures precise and efficient task execution in real time [9].

All modules operate in coordination to complete the overall process. The modular architecture improves flexibility, making it easier to update or modify individual components. This design enhances performance, reliability, and scalability, making the system suitable for real-world applications such as automation, education, and assistive robotics [10].

3.3.1 Speech Recognition Module

In this project, the speech recognition module is used to capture and process voice commands given by the user. A microphone serves as the input device to capture the user's spoken commands. The captured audio signal is then processed using speech recognition techniques, where it is converted into text using Python-based libraries. This allows the system to understand the user's command in a simple and natural way [5].

The module works in real time, enabling quick response to user input. It identifies keywords such as object names (for example, red, green, or blue) and maps them to specific actions. To improve accuracy, noise handling techniques are applied to reduce the effect of background disturbances. Despite these improvements, factors such as accents and pronunciation differences may still affect performance. Therefore, validation is performed before executing commands to ensure correctness [2].

Overall, this module plays an important role in enabling natural human-robot interaction by replacing traditional control methods with voice-based input.

In addition to basic command recognition, the speech module is designed to handle continuous user interaction. The system listens for commands in a loop and processes them dynamically, allowing users to give instructions at any time. This improves usability and makes the system more interactive.

To enhance performance, the system adapts to surrounding noise conditions before capturing input. This allows it to minimize background interference and achieve better recognition accuracy.

Additionally, the module can be expanded to support multiple commands and different languages, this enhances the system's flexibility and makes it suitable for a broader range of users.

3.3.2 Vision-Based Object Detection Module

In this project, the vision-based object detection module is designed to identify and locate objects within the robot's workspace. A camera continuously captures real-time images, which are processed using OpenCV. These images are converted into the HSV color space to achieve better detection performance under varying lighting conditions [6].

The system applies color segmentation to detect objects based on predefined color ranges. After segmentation, contour detection methods are used to outline object boundaries. The centroid of each detected object is then computed to determine its position within the image frame. This positional information is crucial for guiding the robotic arm accurately to the target location [7].

While the system performs effectively in controlled environments, its accuracy can be influenced by factors such as lighting variations, shadows, and complex backgrounds. Nevertheless, the use of the HSV color space enhances robustness and supports reliable performance in real-time scenarios.

To further improve detection quality, the vision module includes preprocessing steps before analysis. Techniques such as image smoothing and noise reduction are applied to enhance image clarity, leading to more accurate detection results.

Additionally, the system continuously observes the workspace, enabling it to detect any changes in object position. This dynamic tracking capability makes the system well-suited for real-time applications where objects may move or shift over time.

3.3.3 Command Interpretation Module

In this project, the command interpretation module serves as a bridge between the input modules and the robotic control system. It receives the text generated by the speech recognition module and analyzes it to determine the intended action. The module extracts key instructions and correlates them with the objects identified by the vision module to ensure correct execution [3].

This module ensures that the system performs the correct action by verifying both the user command and the object detection result. For example, if the user requests a specific object, the system checks whether that object is present in the workspace before executing the command. This validation step reduces errors and improves system safety.

The command interpretation module also handles cases where the input is unclear or invalid. In such situations, the system may ignore the command or request a new input. This improves reliability and ensures smooth system operation.

The command interpretation module also contributes to decision-making by evaluating both the input data and the current system conditions. It ensures that commands are carried out only when all necessary requirements are met. This approach minimizes the likelihood of incorrect operations and enhances the overall reliability of the system.

Furthermore, the module can be extended to support more complex commands, such as combining multiple instructions in a single input. This allows the system to perform advanced tasks and improves user interaction.

3.3.4 Robot Control Module

In this project, the robotic control module is responsible for executing the final task based on the processed input. It receives the object position from the vision module and the corresponding command from the interpretation module. Using this information, the system generates movement instructions for the robotic arm [9].

The robotic arm moves to the detected object location using predefined coordinates. A suction gripper is used to pick the object, and the robot then places it at a specified location. The entire process is performed in real time, ensuring efficient and smooth operation.

To maintain safety, the system incorporates constraints such as predefined movement limits and regulated speed control. These measures ensure that the robot operates within its safe range and avoids unintended motions. Additionally, the module is designed to handle errors, including unsuccessful object detection or invalid commands, thereby maintaining stable and reliable performance.

Overall, the robotic control module completes the system by converting processed data into physical actions, enabling accurate and reliable robotic operations.

The robotic control module includes safety mechanisms to prevent unexpected behavior. It ensures that the robot operates within predefined limits and avoids collisions with surrounding objects. This improves system safety and reliability.

In addition, the module supports smooth motion planning, where the robot moves in a controlled and stable manner. This reduces sudden movements and improves the precision of operations. The system can also be enhanced in the future by adding advanced control algorithms to improve efficiency and performance.

3.4 System Requirements Specifications

In this project, the system requirements specification defines the necessary conditions for designing and implementing the human-centric robotic control system. It includes both software and hardware requirements, as well as functional and non-functional aspects of the system. These

requirements ensure that the system operates efficiently, reliably, and meets the intended objectives [4].

The proposed system is designed to support real-time processing of speech and visual data while controlling a robotic arm. Therefore, it requires appropriate hardware and software components to handle input processing, image analysis, and robotic execution. The requirements are categorized into requirements analysis, software requirements, functional requirements, non-functional requirements, and hardware requirements.

3.4.1 Requirements Analysis

In this project, requirement analysis is conducted to clearly define system needs and ensure effective integration of all components. The primary goal is to provide a simple and intuitive method for controlling a robotic arm using both speech and vision. To achieve this, the system must efficiently process voice inputs and visual data while maintaining reliable real-time performance.

The system should be capable of capturing voice commands from the user through a microphone and converting them into text using speech recognition techniques. It must also be able to interpret these commands correctly and map them to appropriate actions. At the same time, the system should detect objects in the workspace using a camera and process the captured images using computer vision techniques. This requires the system to handle image processing tasks such as color detection, contour extraction, and position calculation accurately [6].

Another key requirement of the system is the seamless integration of its different modules. The speech recognition, vision, and robotic control modules must communicate efficiently to ensure smooth operation. Proper coordination among these components is essential for correctly identifying the target object and executing the intended action. This integration significantly enhances system reliability and minimizes operational errors [4].

The system is designed to operate with real-time processing capabilities. Users should be able to issue commands and receive responses without noticeable delays. This is particularly important in applications such as automation and object handling, where quick and accurate responses are required. To achieve this, efficient algorithms and optimized processing techniques must be implemented to maintain low latency [7].

User requirements are an important consideration in the design process. By using straightforward voice commands, users can control the robot without the need for complex instructions, thereby improving accessibility and usability [1].

Additionally, safety and accuracy are critical aspects of the system. Validation mechanisms must be in place to ensure that commands are executed only when the correct object is detected. This

helps prevent incorrect or unsafe actions. Furthermore, the robotic arm should operate within defined limits to avoid potential damage or accidents during operation [3].

Overall, the requirements analysis helps in defining the structure and functionality of the system. It ensures that the system is designed to be efficient, reliable, user-friendly, and suitable for real-world applications.

3.4.2 Software Requirements

In this project, the software requirements include all the tools, programming languages, and libraries needed to develop and run the human-centric robotic control system. The system is implemented using Python, which is widely used for its simplicity, flexibility, and strong support for various libraries related to artificial intelligence, computer vision, and automation [5].

The vision-based object detection module is developed using the OpenCV library. OpenCV provides efficient functions for image processing tasks such as color conversion, filtering, segmentation, and contour detection. It plays a major role in detecting objects in real time and determining their position within the workspace [6].

For speech recognition, Python-based speech processing libraries are used to capture and convert audio input into text. These libraries allow the system to recognize voice commands and interpret user instructions. Additional libraries such as PyAudio are used to handle microphone input, enabling real-time speech capture and processing.

The system also uses NumPy for numerical operations and matrix calculations. NumPy is essential for handling data related to image processing and coordinate transformation. It helps in performing efficient mathematical computations required for tasks such as homography mapping and coordinate conversion [7].

To control the robotic arm, specific libraries related to the Dobot Magician are used. These libraries provide functions to send movement commands, control speed, and operate the suction gripper. They enable communication between the software and the robotic hardware, ensuring accurate execution of tasks [9].

The system is designed to operate on platforms such as Windows or Linux, which provide a stable environment for running Python applications and supporting the required libraries. Development can be carried out using an integrated development environment (IDE) like Visual Studio Code or PyCharm, which facilitate coding, debugging, and testing.

Overall, the chosen software tools and libraries create a comprehensive environment for implementing speech recognition, image processing, and robotic control. These components enable efficient system performance, support real-time execution, and allow for easy modification or future expansion [4].

3.4.3 Functional Requirements

In this project, the functional requirements define the key operations that the system must perform to achieve its objectives. These requirements describe how the system processes user input, detects objects, and controls the robotic arm to complete tasks accurately and efficiently.

The system must be able to capture voice input from the user using a microphone and convert it into text using speech recognition techniques. It should correctly interpret the command and identify the required action, such as selecting a specific object based on its color. This facilitates a more natural and user-friendly interaction between the user and the robotic system [5].

The system must also be capable of detecting objects in real time using a camera. The vision module should process captured images, apply color segmentation techniques, and identify objects based on predefined color ranges. It should also calculate the position of the detected object using contour detection and centroid estimation methods [6][7].

Another important requirement is coordinate transformation. The system must convert image coordinates into real-world coordinates using mapping techniques such as homography. This ensures that the robotic arm can accurately reach the detected object and perform the required action [8].

The system should include a command validation mechanism. Before executing any action, it must verify both the recognized speech command and the detected object. This validation step helps in reducing errors and ensures that only correct and safe operations are performed [3].

The robotic control module must generate movement instructions for the robotic arm based on the processed data. The robot should be able to move to the target location, pick the object using a suction gripper, and place it at a predefined position. All these operations should be performed in real time with minimal delay [9].

Additionally, the system should handle invalid or unclear inputs effectively. If the command is not recognized or the object is not detected, the system should avoid performing any action and wait for a valid input. This improves system reliability and prevents incorrect operations.

Overall, the functional requirements ensure that the system performs all necessary tasks efficiently, accurately, and safely, enabling effective human–robot interaction in real-world applications.

3.4.4 Non-Functional Requirements

In this project, non-functional requirements define the quality standards and performance expectations that the system must meet. These requirements ensure that the system performs efficiently and reliably, while remaining user-friendly and maintaining high levels of safety and accuracy.

One of the most critical non-functional requirements is real-time performance. The system must process both speech input and visual data with minimal delay, allowing the robotic arm to respond promptly to user commands. Any significant delay can reduce system efficiency and negatively impact the user experience, particularly in real-time applications such as object handling and automation [7].

Another key requirement is **reliability**. The system should consistently perform its tasks without failure under different operating conditions. It must accurately recognize speech commands and detect objects even in the presence of minor environmental variations such as noise or lighting changes. Reliable operation is essential to ensure correct and safe execution of robotic actions [6].

The system must also be **user-friendly**. It should provide an easy and intuitive interface that allows users to interact with the robot using simple voice commands. The system should not require advanced technical knowledge, making it accessible to a wide range of users, including beginners [1].

Accuracy is another important requirement. The system should correctly interpret user commands and accurately detect object positions. High accuracy in both speech recognition and vision processing is necessary to ensure precise robotic movements and successful task execution [5].

The system should also be **scalable and flexible**. It should allow future improvements such as adding new commands, supporting additional object types, or integrating advanced technologies like machine learning. A flexible design ensures that the system can adapt to new requirements without major modifications [4].

Safety is a critical requirement in robotic systems. The robotic arm should operate within predefined limits to avoid collisions or damage. The system must ensure that commands are validated before execution to prevent unintended actions. Safety mechanisms help protect both the user and the equipment during operation [3].

Another important aspect is **maintainability**. The system should be easy to maintain, debug, and update. A modular design helps in isolating issues and improving system performance without affecting the entire system.

Finally, the system should be **efficient in resource usage**. It should use available hardware resources such as memory and processing power effectively to maintain smooth operation without unnecessary load.

Overall, these non-functional requirements ensure that the system is reliable, efficient, safe, and easy to use, making it suitable for real-world applications.

3.4.5 Hardware Requirements

In this project, the hardware requirements include the physical components necessary for implementing the human-centric robotic control system. These components are selected to ensure efficient performance of speech recognition, image processing, and robotic operations in real time.

- **Processor:** An Intel Core i5 or higher processor is recommended to handle computational tasks such as speech processing and image analysis efficiently. A higher-performance processor ensures smooth execution of real-time operations without delays.
- **RAM:** A minimum of 8 GB RAM is required to support multitasking and allow multiple modules, such as speech recognition and vision processing, to run simultaneously without performance issues.
- **Storage:** At least 128 GB of storage is needed to install required software, libraries, and to store project-related data. Sufficient storage ensures smooth operation and easy management of files.
- **Camera:** A high-resolution webcam is used to capture clear and detailed images, which are essential for accurate object detection. Improved image quality enhances the performance of vision-based algorithms, especially under different lighting conditions.
- **Microphone:** A reliable microphone is required to capture voice commands clearly. It should be capable of minimizing background noise to improve the accuracy of speech recognition.
- **Robotic Arm:** A robotic arm such as the Dobot Magician is utilized to carry out pick-and-place operations. It offers precise movement and supports programmable control for efficient task execution.
- **End Effector:** A suction gripper is employed as the end effector to handle objects. It provides stable gripping and improves the efficiency of pick-and-place operations.
- **Connectivity:** Communication through USB or serial interfaces is required to connect the computer with the robotic arm. This enables smooth and reliable transmission of control commands from the system to the robot.

Overall, these hardware components function together to enable accurate, efficient, and real-time operation of the system.

3.5 Introduction to UML (Unified Modeling Language)

In this project, Unified Modeling Language (UML) is adopted as a standard method for visualizing and designing the system architecture. It provides graphical representations that make it easier to understand both the structure and behavior of complex systems. By breaking the system into different diagrams, UML simplifies the design process and supports better analysis and development.

UML is commonly used in software engineering to model object-oriented systems. It helps in illustrating system components, their relationships, and the way they interact with each other. Using UML diagrams allows developers to clearly convey system design to team members and stakeholders, reducing misunderstandings and improving overall collaboration.

In this project, UML diagrams are used to represent the working of the human-centric robotic control system. These diagrams illustrate how different modules, such as speech recognition, vision-based object detection, command interpretation, and robotic control, interact with each other. They also help in understanding the flow of data and the sequence of operations within the system.

Overall, UML plays an important role in improving system clarity, design efficiency, and documentation, making it easier to develop and maintain the proposed system.

3.5.1 Class Diagram

In this project, the class diagram represents the structural design of the system by showing different classes and their relationships. It helps in understanding how various components are organized and interact with each other. The main classes in the system include *SpeechRecognizer*, *CameraModule*, *ObjectDetector*, *CommandProcessor*, and *RobotController*.

Each class includes defined attributes and methods that determine its behavior and functionality. The *SpeechRecognizer* class handles voice input processing, while the *CameraModule* captures real-time images. The *ObjectDetector* class processes images to detect objects based on color. The *CommandProcessor* interprets user commands, and the *RobotController* executes actions using the robotic arm. This diagram supports modular system design and improves code organization [4].

3.5.2 Component Diagram

In this project, the component diagram offers a high-level representation of the system by illustrating its main modules and how they interact with each other. The key components include the speech recognition module, vision processing module, command interpretation module, and robotic control module.

Each component is designed to perform a specific function while communicating with other modules to ensure efficient task execution. The speech module captures user commands, the vision module identifies objects within the workspace, and the control module carries out the required robotic actions. This diagram provides a clear understanding of the system architecture and highlights the dependencies among different modules [4][10].

3.5.3 Deployment Diagram

In this project, the deployment diagram represents the physical implementation of the system by illustrating how hardware components are interconnected. It shows how software modules are deployed across devices such as the computer, microphone, camera, and robotic arm.

The computer functions as the central processing unit, where all system modules are executed and coordinated. The microphone captures voice input, the camera captures visual data, and the robotic arm

executes commands. This diagram helps in understanding the interaction between software and hardware components in real-time operation [9].

3.5.4 Object Diagram

In this project, the object diagram represents specific instances of system components during execution. It shows how objects interact at a particular moment in time.

For example, a voice command object is processed by the speech module, generating a command object. This command is then validated using object detection results before being sent to the robotic controller. This diagram provides a clear understanding of runtime system behavior [3].

3.5.5 Use Case Diagram

In this project, the use case diagram illustrates how users interact with the system. The user is the main actor who provides voice commands to control the robotic arm.

The system processes these commands, detects objects, and performs actions such as picking and placing objects. Key interactions include “Provide Voice Command,” “Detect Object,” and “Control Robot.” This diagram helps in understanding system functionality from a user perspective [1].

3.5.6 Activity Diagram

In this project, the activity diagram illustrates the overall workflow of the system by outlining the sequence of steps involved in task execution. The process begins with capturing a voice command, followed by command processing, object detection, coordinate transformation, and finally robotic execution. Decision nodes are incorporated to verify both the command and detection results before proceeding. This diagram provides a clear visualization of the system flow and helps in improving operational efficiency [7].

3.5.7 Sequence Diagram

In this project, the sequence diagram depicts the interaction between different modules over time, emphasizing the order of operations. The process starts when the user gives a voice command, which is handled by the speech recognition module. The system then activates the vision module to detect and identify the object.

After validation, the processed command is forwarded to the robotic control module for execution. This diagram helps in understanding the sequence and timing of interactions among system components [5][6].

3.5.8 Collaboration Diagram

In this project, the collaboration diagram illustrates how system components communicate with each other. It focuses on message exchange between modules rather than the sequence of operations.

The diagram shows how speech input, vision processing, and robotic control modules interact to complete tasks. It highlights coordination between different components and helps in improving system design clarity [4][10].

3.6 UML Diagrams

In this project, Unified Modeling Language (UML) diagrams are used to represent the structure and working of the system in a clear and understandable manner. UML provides a visual way to describe how different components of the system are organized and how they interact with each other. It helps in simplifying complex system behavior by presenting it in the form of diagrams, which makes analysis and design easier [4].

UML diagrams are useful for visualizing the flow of data, communication between modules, and overall system functionality. They allow developers to understand how different parts of the system are connected and how information moves from one module to another. This improves clarity in system design and helps in identifying potential issues during development.

In this project, UML diagrams are used to represent key processes such as capturing voice input, processing speech commands, detecting objects using computer vision, performing coordinate mapping, and executing actions using the robotic arm. These diagrams show how the system processes user input step by step and converts it into meaningful actions.

Various UML diagrams, including class diagrams, use case diagrams, activity diagrams, and sequence diagrams, are utilized to represent both the structural and behavioral aspects of the system. Structural diagrams describe the system components and their relationships, while behavioral diagrams illustrate how the system functions over time.

Overall, UML diagrams significantly enhance system understanding, design clarity, and documentation. They offer a clear visual representation that helps in communicating ideas, analyzing system functionality, and supporting efficient development processes [10].

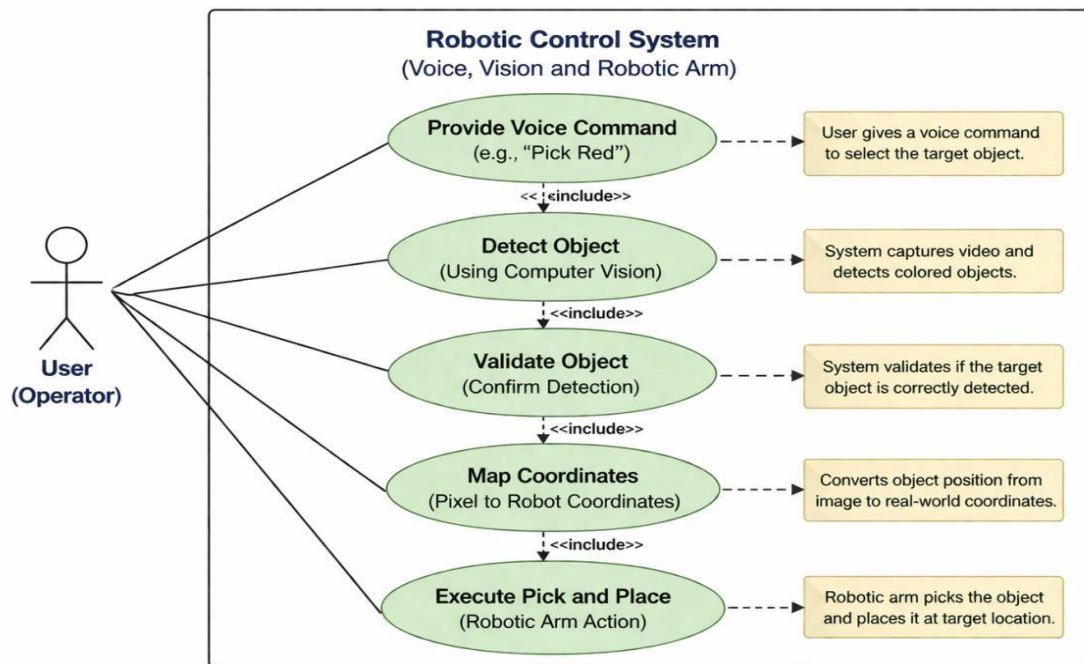


Fig 3.6.1 Use Case Diagram

The use case diagram illustrates how the user interacts with the human-centric robotic control system [1]. In this project, the user acts as the main actor who provides voice commands to control the robotic arm [2]. The process begins with capturing the user's speech through a microphone, which is then converted into text using speech recognition techniques [5].

Based on the recognized command, the system activates the vision module to identify objects present in the workspace using computer vision methods [6]. Once the object is detected, the system performs a validation step to ensure that the correct object has been identified. This verification improves accuracy and prevents incorrect operations [3].

After successful detection, the position of the object is converted from image coordinates into real-world coordinates using mapping techniques such as homography [8]. This transformation is important because it allows the robotic arm to move precisely to the target location.

Once the coordinates are determined, the robotic control module executes the required action, such as picking and placing the object at a predefined position [9]. Each stage of the system is connected, ensuring smooth data flow and reliable operation.

Overall, the use case diagram clearly represents the workflow of the system, showing how speech recognition and vision-based detection are integrated to achieve efficient and accurate robotic control [10].

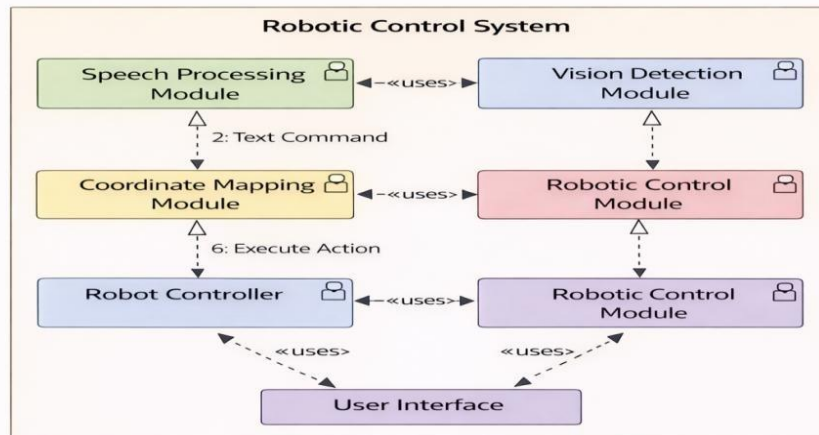


Fig 3.6.2 Robotic Control System

The diagram represents the system architecture of the human-centric robotic control system, highlighting how different functional modules interact with each other [1]. In this project, the system is designed as a combination of multiple interconnected modules that work together to achieve accurate and real-time operation.

The speech processing module is responsible for capturing voice commands from the user and converting them into text using speech recognition techniques [5]. This processed command is then passed to the next stages for further interpretation. The vision detection module plays an important role in identifying objects present in the workspace using computer vision methods [6].

Once an object is detected, the coordinate mapping module converts its position from image coordinates into real-world coordinates. This transformation ensures that the robotic arm can accurately reach the target object [8]. After this, the robotic control module processes both the command and object position to generate appropriate movement instructions for the robot [9].

The robot controller then executes the final operation, such as picking and placing the object, based on the received instructions [9]. The user interface acts as a link between the user and the system, allowing easy interaction and control through voice commands [2].

All modules in the system are connected through “uses” relationships, which indicate communication and dependency between them [4]. Each module depends on the output of the previous module, forming a sequential and coordinated workflow. This structured design ensures that the system operates efficiently and maintains real-time performance.

Overall, the diagram provides a clear representation of how speech recognition, vision-based detection, and robotic control are integrated into a single system, enabling accurate and reliable robotic operations [10].

CHAPTER – 4

SYSTEM IMPLEMENTATION

System implementation defines how the proposed system is developed in terms of both its physical setup and operational workflow. This phase ensures that the system becomes fully functional and capable of performing tasks in real time. It also focuses on verifying that the system meets the required standards of performance, accuracy, and reliability.

In this project, the implementation phase involves integrating multiple modules such as speech recognition, vision-based object detection, coordinate mapping, and robotic control into a single working system. Each module is designed and tested individually before being combined to ensure smooth interaction and efficient data flow between components. This integration allows the system to process user input, analyze the environment, and execute robotic actions in a coordinated manner.

The implementation also considers real-time performance, where the system continuously captures voice commands and visual data, processes them, and controls the robotic arm without noticeable delay. Proper synchronization between modules ensures that the system operates accurately and reliably under different conditions. Overall, the implementation phase transforms the system design into a fully operational solution suitable for real-world applications [4][10].

4.1 Technologies

In this project, several technologies are used to develop and implement the human-centric robotic control system. These technologies work together to enable voice-based interaction, real-time object detection, and accurate robotic control.

Python is used as the main programming language due to its simplicity, flexibility, and strong support for libraries related to speech processing, computer vision, and robotics. Speech recognition libraries are used to capture and convert voice input into text, allowing users to control the system using natural language commands [5].

OpenCV is used for processing visual input from the camera. It enables real-time image analysis, object detection, and feature extraction. HSV color segmentation techniques are applied to identify objects based on their color, providing reliable detection under varying lighting conditions [6][7].

Homography mapping is utilized to convert image-based coordinates into real-world coordinates. This transformation allows the robotic arm to accurately determine object positions and interact with them effectively within the workspace [8].

Dobot control libraries are used to communicate with and control the robotic arm. These libraries enable the system to send movement commands, control speed, and perform pick-and-place

operations efficiently. The integration of these technologies ensures smooth coordination between input, processing, and execution stages [9].

Overall, the combination of these technologies enables the system to function effectively by integrating speech recognition, computer vision, and robotic control into a unified framework. This results in a reliable, efficient, and user-friendly system suitable for real-time applications [10].

4.1.1 Python

In this project, Python is used as the primary programming language for developing the human-centric robotic control system. Python is a high-level language known for its simplicity, readability, and ease of use, making it suitable for both beginners and advanced developers. Due to these features, it is widely used in fields such as robotics, automation, and computer vision [5].

Python provides a flexible and efficient environment for integrating different components of the system. In this project, it is used to implement speech recognition, vision-based object detection, and robotic control within a single framework. The availability of powerful libraries such as SpeechRecognition for audio processing and OpenCV for image processing makes Python highly suitable for handling both voice and visual data [5][6].

Another key advantage of Python is that it is an interpreted language, allowing code to be executed step by step. This feature makes debugging easier and speeds up the overall development process.

Python also supports object-oriented programming principles such as classes and objects, which help in building a modular and well-organized system. With this modular approach, components like speech processing, vision detection, and robotic control can operate independently while still working together seamlessly [4].

In addition, Python offers strong cross-platform compatibility. Applications developed using Python can run on operating systems such as Windows, Linux, and macOS with minimal modifications. This makes it an ideal choice for creating portable robotic systems that can be deployed across different environments.

In addition, Python provides automatic memory management and a large collection of built-in libraries, which reduce development effort and improve efficiency. Its strong community support ensures continuous updates, extensive documentation, and availability of resources, making it easier to develop and maintain complex applications.

Overall, Python plays a vital role in this project by providing a reliable and efficient platform for integrating speech recognition, computer vision, and robotic control. It enables smooth development and supports real-time operation of the system, making it an essential component of the proposed solution [10].

4.1.2 Speech Recognition

In this project, speech recognition serves as a core technology that enables the system to transform spoken language into text, allowing users to control the robot through natural voice commands. This approach enhances human–computer interaction by reducing reliance on conventional input devices such as keyboards or controllers, thereby improving both usability and efficiency [5].

The system utilizes a microphone to capture the user’s voice input. The recorded audio is processed using Python-based speech recognition libraries, where it is converted into a digital signal and analyzed to extract meaningful text. This processed text is then examined to identify specific keywords associated with predefined actions, such as selecting objects based on their color [5].

A command-based approach is followed in this system, where each recognized voice command is mapped to a specific robotic action. For example, commands such as “pick red,” “pick green,” or “pick blue” are interpreted and converted into instructions for the robotic arm. This mapping ensures that the system responds correctly to user input and performs the desired task efficiently.

One of the major advantages of speech recognition is that it enables contactless interaction. Users can control the robotic system without physically touching any device, which improves safety and convenience. This feature is particularly useful in environments such as industrial automation and healthcare, where minimizing physical contact is important [2].

To improve recognition accuracy, the system may include preprocessing techniques such as noise reduction and filtering to reduce the effect of background disturbances. Additionally, the system is designed to recognize commands within a limited vocabulary, which helps reduce ambiguity and improves performance. However, factors such as variations in accent, pronunciation, and environmental noise can still affect the accuracy of speech recognition [5].

The integration of speech recognition with vision-based object detection further enhances system reliability. While the speech module provides the command, the vision module verifies the presence of the object before execution. This combined approach ensures that actions are performed accurately and safely [3].

Overall, speech recognition provides a simple, efficient, and user-friendly interface for controlling the robotic system. Its ability to interpret natural voice commands and convert them into actions makes it an essential component of the proposed human-centric robotic control system.

4.1.3 OpenCV

In this project, OpenCV is utilized as the primary tool for implementing vision-based processing. It is a widely recognized open-source library developed for real-time image processing and computer vision tasks. OpenCV offers a broad range of functions that enable efficient acquisition,

processing, and analysis of images and video, making it highly suitable for robotics and automation applications [6].

The system utilizes OpenCV to acquire live video frames from a webcam. These frames are processed to detect objects within the workspace. Before detection, the images undergo preprocessing steps such as resizing and filtering to enhance image quality. These operations help reduce noise and highlight important features, thereby improving the accuracy of object detection [6].

A key operation performed using OpenCV is color space conversion. The system converts images from RGB format to HSV color space, which improves the reliability of color-based detection under varying lighting conditions. By separating color information from brightness, this approach allows the system to identify objects more effectively.

After color segmentation, OpenCV is used to perform contour detection. This helps in identifying object boundaries and extracting properties such as shape and area. The centroid of each detected object is then calculated to determine its position within the image frame. This positional data is essential for guiding the robotic arm accurately during task execution [7].

Another important advantage of OpenCV is its ability to support real-time processing. The system can continuously capture and analyze video frames with minimal delay, enabling smooth interaction between the user and the robot. Visual outputs such as contours, bounding boxes, and labels can also be displayed, allowing easy monitoring of system performance.

OpenCV can be seamlessly integrated with other modules, including speech recognition and robotic control. Its compatibility with Python simplifies development and maintenance, while its optimized algorithms ensure efficient performance even on standard computing systems.

Overall, OpenCV plays a crucial role in the vision module of this project. It enables accurate and real-time object detection, supports efficient image processing, and links visual input with robotic actions, thereby enhancing the overall reliability and effectiveness of the system [10].

4.1.4 HSV Color Segmentation

In this project, HSV (Hue, Saturation, Value) color segmentation is used as an effective technique for detecting objects based on their color properties. Unlike the RGB color model, where color and brightness are combined, HSV separates these components into three parameters: hue (color type), saturation (color intensity), and value (brightness). This separation makes HSV more suitable for object detection, especially under varying lighting conditions [6].

The system captures images using a camera and converts them from RGB color space into HSV color space using OpenCV. This conversion improves the system's ability to distinguish between colors, even when lighting conditions are not consistent. By focusing on hue values, the system

can detect objects more reliably while reducing the influence of shadows and brightness variations [6].

After conversion, predefined color ranges are applied to isolate specific objects. For example, separate threshold values are defined for colors such as red, green, and blue. These thresholds are used to generate a binary mask, where the selected color is highlighted and other regions are ignored. This process allows the system to clearly identify the target object within the workspace [7].

To enhance detection accuracy, additional image processing techniques such as noise filtering and morphological operations are applied. These methods help eliminate unwanted noise and refine the segmented regions. Once the object is clearly isolated, contour detection is used to determine its shape and calculate its position within the image frame.

One of the key advantages of HSV color segmentation is its simplicity and computational efficiency. It does not rely on complex models or large datasets, making it well-suited for real-time applications. It performs particularly well in controlled environments where objects can be easily distinguished based on their color.

However, this technique also has certain limitations. Its effectiveness may decrease under extreme lighting conditions or when objects have colors similar to the background. To address this issue, careful calibration of threshold values is necessary to ensure consistent detection performance.

Overall, HSV color segmentation serves as a practical and dependable approach for object detection in this project. It allows accurate identification of colored objects and supports precise robotic operations by providing stable and efficient visual input [10].

4.1.5 Homography Mapping

In this project, homography mapping is used as an important technique to convert coordinates from the image plane to the real-world workspace. It is a mathematical method that establishes a relationship between two planes, allowing transformation from pixel coordinates to actual physical coordinates required by the robotic system [8].

In vision-based robotic applications, the camera captures object positions in terms of image pixels. However, the robotic arm cannot directly use these pixel values to perform actions. Therefore, homography mapping is applied to convert image coordinates into real-world coordinates so that the robot can accurately interact with objects in its workspace. This process forms a critical link between the vision module and the robotic control system [8].

The mapping process begins with a calibration step. During calibration, a set of known reference points is selected from both the image and the physical workspace. Once this matrix is obtained,

it can be used to convert any point from the image into its corresponding real-world position. This ensures that the robotic arm moves precisely to the detected object location.

Accurate calibration is essential for effective homography mapping. Any errors in selecting reference points or computing the transformation matrix can lead to incorrect positioning of the robotic arm. Therefore, careful calibration and validation are required to maintain system accuracy. Factors such as camera alignment, viewing angle, and distance from the workspace must also be properly adjusted to achieve reliable results [9].

By accurately transforming coordinates, the system ensures that the robotic arm aligns correctly with the actual position of objects. This is especially important for tasks such as pick-and-place operations, where even small positional errors can affect performance.

Another advantage of this technique is its efficiency. Once the transformation matrix is calculated, coordinate conversion can be performed quickly, making it suitable for real-time applications. This allows the system to continuously update object positions and respond dynamically to changes in the environment.

Overall, homography mapping plays a crucial role in the proposed system by enabling accurate coordination between the vision module and the robotic arm. It ensures precise object localization and improves the overall performance and reliability of the system in real-time robotic applications [10].

4.1.6 Robot Control (Dobot)

In this project, robot control is a key component that enables the system to convert processed input data into physical actions. The Dobot Magician robotic arm is used as the main actuator to perform operations such as picking and placing objects. The robot control module is responsible for receiving commands from the system and translating them into precise movements of the robotic arm [9].

The Dobot Magician is a versatile robotic arm that supports programmable control through software interfaces. It is capable of performing movements along multiple axes, handling objects using a suction gripper, and executing repetitive tasks with good precision. The robot is connected to the computer system through a communication interface, typically using a USB connection, which allows real-time exchange of data between software and hardware components.

In this system, the robot control module receives processed information from the command interpretation and coordinate mapping modules. Based on the detected position of the object, movement instructions are generated to guide the robotic arm accurately. The robot first moves to a position above the object, then lowers its arm to pick the object using the suction gripper. After successfully grasping the object, it lifts the object and moves it to a predefined location for placement.

The control mechanism ensures that all movements are performed within safe operating limits to prevent errors or collisions. It also maintains smooth and stable motion by controlling speed and direction during operation. The integration of the robot with speech and vision modules allows coordinated operation, where actions are executed only after proper validation of both command and object detection [4].

One of the main advantages of using the Dobot Magician is its ease of integration and high level of accuracy. It supports real-time control and can be programmed using Python libraries, making it suitable for both educational and automation purposes. The robot provides consistent and reliable performance during pick-and-place operations [9].

Overall, the robot control module plays an important role in converting processed system data into real-world actions. It ensures that the robotic arm operates efficiently, accurately, and safely, completing tasks based on user commands and visual input.

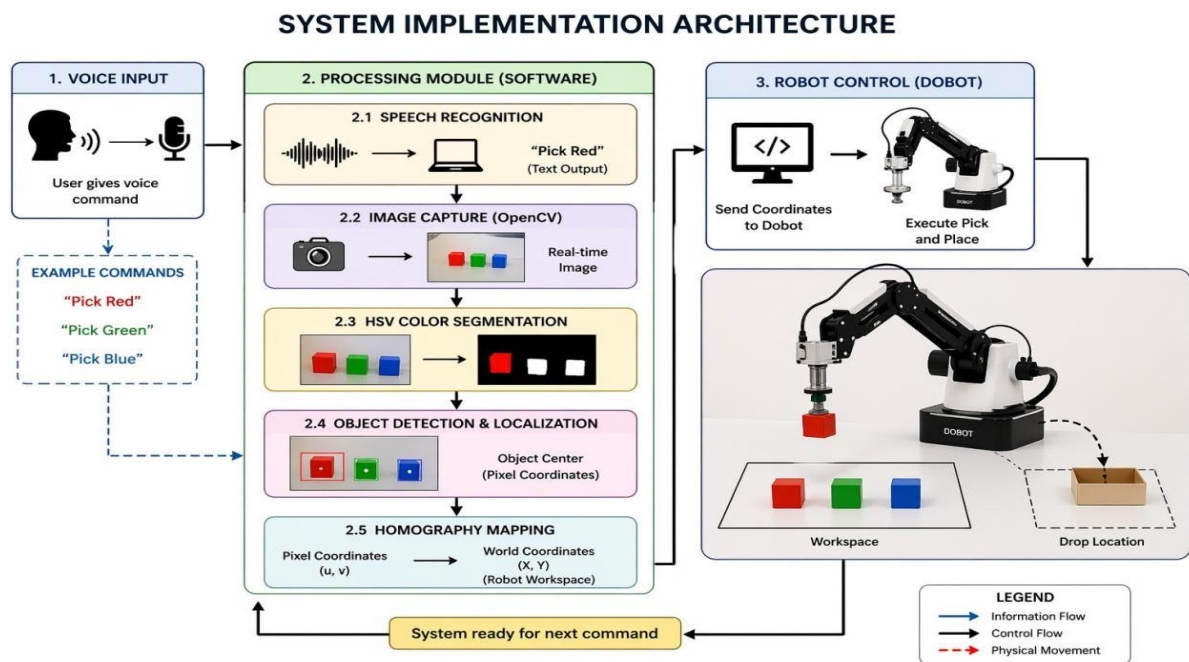


Fig 4.1.6 System Implementation Architectue

4.2 Code

```
import cv2
import numpy as np
import time
import speech_recognition as sr
from pydobot import Dobot

# =====
# DOBOT CONFIG
# =====

device = Dobot(port="COM3", verbose=False)
device.speed(50)

HOME_Z = 40
PICK_Z = -43

DROP_POSITIONS =
    { "RED": (285.7, -156.2, 40),
      "GREEN": (213.8, -206.9, 40),
      "BLUE": (132.1, -192.0, 40),
    }

SAFE_MIN_X = 150
SAFE_MAX_X = 300
SAFE_MIN_Y = -60
SAFE_MAX_Y = 250

# Fine tuning offsets
FINE_OFFSET_X = -3
FINE_OFFSET_Y = 30

# =====
# LOAD HOMOGRAPHY
# =====

H = np.load("homography.npy")

# =====
# VOICE RECOGNITION
# =====
```

```
def listen_command():
    r = sr.Recognizer()
    with sr.Microphone() as source:
        print("\nSay: Pick Red / Pick Green / Pick Blue")
        r.adjust_for_ambient_noise(source)
        audio = r.listen(source)

    try:
        command = r.recognize_google(audio)
        command = command.lower()
        print("You said:", command)

        if "red" in command:
            return "RED"
        elif "green" in command:
            return "GREEN"
        elif "blue" in command:
            return "BLUE"
        else:
            return None

    except:
        print("Voice not recognized")
        return None

# =====
# UTILS
# =====

def clamp(val, minv, maxv):
    return max(min(val, maxv), minv)

def pixel_to_robot(px, py):
    pts = np.array([[px, py]], dtype=np.float32)
    robot_pts = cv2.perspectiveTransform(pts, H)

    x = float(robot_pts[0][0][0]) + FINE_OFFSET_X
    y = float(robot_pts[0][0][1]) + FINE_OFFSET_Y

    x = clamp(x, SAFE_MIN_X, SAFE_MAX_X)
    y = clamp(y, SAFE_MIN_Y, SAFE_MAX_Y)
```

```
    return x, y

# =====
# ROBOT PICK FUNCTION
# =====

def move_robot(robot_x, robot_y, color):
    drop_x, drop_y, drop_z = DROP_POSITIONS[color]

    print(f"\nPicking {color} at {robot_x:.1f}, {robot_y:.1f}")

    try:
        device.move_to(robot_x, robot_y, HOME_Z, 0, wait=True)
        device.move_to(robot_x, robot_y, PICK_Z, 0, wait=True)

        device.suck(True)
        time.sleep(1)

        device.move_to(robot_x, robot_y, HOME_Z, 0, wait=True)
        device.move_to(drop_x, drop_y, HOME_Z, 0, wait=True)
        device.move_to(drop_x, drop_y, drop_z, 0, wait=True)

        device.suck(False)
        time.sleep(1)

        device.move_to(drop_x, drop_y, HOME_Z, 0, wait=True)

    except Exception as e:
        print("Robot error:", e)

# =====
# COLOR RANGES
# =====

color_ranges =
{ "RED": [
    (np.array([0, 120, 70]), np.array([10, 255, 255])),
    (np.array([170, 120, 70]), np.array([180, 255, 255]))
], "GREEN":
[
    (np.array([35, 80, 80]), np.array([85, 255, 255]))
```

```
], "BLUE":
[
    (np.array([90, 80, 80]), np.array([130, 255, 255]))
]
}

# =====
# CAMERA
# =====

cap = cv2.VideoCapture(0, cv2.CAP_DSHOW)

if not cap.isOpened():
    print("Camera failed")
    exit()

print("Camera started")

# =====
# MAIN LOOP
# =====

while True:
    selected_color = listen_command()

    if selected_color is None:
        continue

    print("Searching for:", selected_color)

    while True:
        ret, frame = cap.read()
        if not ret:
            continue

        frame = cv2.resize(frame, (640, 480))
        display = frame.copy()

        hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV)

        ranges = color_ranges[selected_color]
        mask = None
```

```
for lower, upper in ranges:
    m = cv2.inRange(hsv, lower, upper)
    mask = m if mask is None else mask + m

# Morphology
kernel = np.ones((5, 5), np.uint8)
mask = cv2.morphologyEx(mask, cv2.MORPH_CLOSE, kernel)

contours, _ = cv2.findContours(mask, cv2.RETR_EXTERNAL,
cv2.CHAIN_APPROX_SIMPLE)

for cnt in contours:
    area = cv2.contourArea(cnt)
    if area < 2000:
        continue

    x, y, w, h = cv2.boundingRect(cnt)

    if x < 80:
        continue

    M = cv2.moments(cnt)
    if M["m00"] == 0:
        continue

    cx = int(M["m10"] / M["m00"])
    cy = int(M["m01"] / M["m00"])

    # Draw
    cv2.rectangle(display, (x, y), (x+w, y+h), (0, 255, 0), 2)
    cv2.circle(display, (cx, cy), 6, (0, 0, 255), -1)

    cv2.putText(display, selected_color, (x, y-10),
                cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0,255,255), 2)

    robot_x, robot_y = pixel_to_robot(cx, cy)

    cv2.putText(display,
                f"Robot ({robot_x:.1f},{robot_y:.1f})",
                (x, y+h+20),
                cv2.FONT_HERSHEY_SIMPLEX,
```

```
        0.5,  
        (255,255,255),  
        1)  
  
    cv2.imshow("Robot Vision", display)  
  
    move_robot(robot_x, robot_y, selected_color)  
    break  
  
    cv2.imshow("Robot Vision", display)  
  
    if cv2.waitKey(1) & 0xFF == ord('q'):  
        break  
  
    break  
  
cap.release()  
cv2.destroyAllWindows()  
device.close()
```

CHAPTER – 5

SYSTEM TESTING

5.1 Test Case Description

System testing plays a crucial role in the development of the proposed human-centric robotic control system, as it ensures that the system functions correctly according to the defined requirements. In this phase, the overall system is evaluated to verify that all components work together smoothly and produce the expected results [10].

The testing process includes evaluating different modules such as speech recognition, vision-based object detection, coordinate mapping, and robotic control. Each module is first tested individually to confirm its functionality, and then all modules are tested together in an integrated environment. This approach helps in ensuring proper communication between components and accurate system operation [4].

Test cases are carefully designed to assess the system's ability to recognize voice commands, detect objects correctly, and perform pick-and-place operations efficiently. These test cases include both normal and edge conditions to validate system performance. The system is also tested under different environmental conditions, such as variations in lighting and background noise, to evaluate its robustness and adaptability [6].

For each test case, expected results are defined in advance, and the actual outputs obtained during testing are compared with these expected values. This comparison helps in identifying errors, measuring system accuracy, and ensuring that the system meets performance requirements [5].

Overall, system testing confirms that the proposed system operates reliably, performs tasks accurately, and is suitable for real-time applications in robotic control.

5.2 Types of Testing

In this project, different types of testing are performed to ensure that the system operates accurately, reliably, and efficiently under various conditions. These testing methods help in verifying individual components as well as the overall system performance.

Unit Testing:

In unit testing, each module is evaluated independently to ensure that it functions correctly. Modules such as speech recognition, vision-based object detection, and robotic control are evaluated separately to ensure that they perform their intended tasks correctly [5][6].

Integration Testing:

Integration testing is carried out by combining all modules and testing them together. This ensures

proper communication and data flow between different components such as speech processing, vision detection, and robotic control. It helps in identifying issues related to module interaction [4].

System Testing:

System testing evaluates the complete system as a whole. It verifies whether all components work together correctly and whether the system meets its overall objectives. This type of testing ensures that the system performs as expected in real-world scenarios [10].

Performance Testing:

Performance testing is used to measure system efficiency, including response time, processing speed, and real-time execution. It ensures that the system responds quickly to user commands and performs tasks without noticeable delay [7].

User Acceptance Testing:

In this testing, the system is evaluated by users to ensure that it meets their requirements. It focuses on usability, ease of interaction, and overall user satisfaction. This testing ensures that the system is user-friendly and practical for real-world use [1].

In addition to these basic testing methods, further validation is performed to improve system robustness and stability. The system is tested in real-time environments with variations in lighting conditions, background noise, and object placement. These tests help identify potential issues and enhance system performance.

Regression Testing:

Regression testing is conducted after making changes or updates to the system. It ensures that previously working functionalities remain unaffected and that the system continues to operate correctly [4].

Stress Testing:

Stress testing evaluates system performance under extreme conditions, such as continuous operation or handling multiple commands. This ensures that the system remains stable and does not fail under heavy usage [7].

Accuracy Testing:

Accuracy testing measures how correctly the system interprets speech commands and detects objects. High accuracy is essential for reliable operation and successful task execution [5][6].

Safety Testing:

Safety testing ensures that the robotic arm operates within defined limits and avoids unintended movements or collisions. It verifies that the system can handle unexpected inputs and errors safely [9].

Usability Testing:

Usability testing evaluates how easily users can interact with and operate the system. It ensures that users can interact with the system comfortably without requiring technical expertise. This improves accessibility and overall user experience [1].

Overall, these testing methods ensure that the system is not only functional but also reliable, safe, and efficient under different operating conditions. They play a crucial role in validating the performance and stability of the proposed system.

5.3 Test Cases

Test Case Tables for Speech Recognition

Positive Test Cases**Table 5.3.1 Positive Test Cases for Speech Recognition**

Function Name	Expected Output	Executed Output	Result Status
Say "Pick Red"	Red command should be detect	Red command detected correct	Successful
Say "Pick Green"	Green command should be dete	Green command detected corre	Successful
Say "Pick Blue"	Blue command should be detect	Blue command detected correc	Successful
Clear voice input	Accurate recognition	Command recognized correctly	Successful

Negative Test Cases**Table 5.3.2 Negative Test Cases for Speech Recognition**

Function Name	Expected Output	Executed Output	Result Status
No voice input	No action should be performed	No action executed	Successful
Background noise	Reduced accuracy	Command not recognized	Successful
Invalid command	System should ignore	No command executed	Successful
Unclear speech	No recognition	Command ignored	Successful

Test Case Tables for Object Detection

Positive Test Cases

Table 5.3.3 Positive Test Cases for Object Detection

Function Name	Expected Output	Executed Output	Result Status
Detect Red Object	Red object should be detected	Red object detected correctly	Successful
Detect Green Object	Green object should be detected	Green object detected correctly	Successful
Detect Blue Object	Blue object should be detected	Blue object detected correctly	Successful
Proper lighting	Accurate detection	Object detected correctly	Successful

Negative Test Cases

Table 5.3.4 Negative Test Cases for Object Detection

Function Name	Expected Output	Executed Output	Result Status
No object present	No detection	No object detected	Successful
Low lighting	Detection may fail	Object not detected properly	Successful
Similar colors	Avoid wrong detection	No incorrect detection	Successful
Partial object	Ignore detection	No detection triggered	Successful

Test Case Tables for Coordinate Mapping

Positive Test Cases

Table 5.3.5 Positive Test Cases for Coordinate Mapping

Function Name	Expected Output	Executed Output	Result Status
Pixel to robot conversio	Accurate coordinate mappin	Coordinates mapped correctl	Successful
Valid object position	Correct robot position	Robot reaches correct locatio	Successful
Calibration applied	Accurate mapping	Mapping correct	Successful

Negative Test Cases

Table 5.3.6 Negative Test Cases for Coordinate Mapping

Function Name	Expected Output	Executed Output	Result Status
Incorrect calibration	Mapping error	Slight position error	Successful
Out of bounds values	Limit position	Values restricted	Successful
Invalid input	Ignore mapping	No movement	Successful

Test Case Tables for Robot Control

Positive Test Cases

Table 5.3.7 Positive Test Cases for Robot Control

Function Name	Expected Output	Executed Output	Result Status
Pick command	Object should be picked	Object picked correctly	Successful
Place command	Object should be placed	Object placed correctly	Successful
Suction ON	Activate suction	Suction activated	Successful
Suction OFF	Deactivate suction	Suction deactivated	Successful

Negative Test Cases

Table 5.3.8 Negative Test Cases for Robot Control

Function Name	Expected Output	Executed Output	Result Status
Invalid command	Ignore command	No movement	Successful
Out of range position	Limit movement	Movement restricted	Successful
Communication failure	No execution	Error handled	Successful

5.4 Results

The results obtained from system testing show that the proposed human-centric robotic control system operates effectively by integrating speech recognition, vision-based object detection, and robotic control into a single framework [10]. The system demonstrates the ability to interpret user commands such as “pick red,” “pick green,” and “pick blue” with good accuracy under normal conditions. This enables a natural and intuitive interaction between the user and the robotic system [5]. In addition, the system processes voice input in real time, ensuring quick response and smooth operation.

The vision-based object detection module performs reliably by identifying objects based on color using HSV color segmentation techniques. The system is able to detect objects accurately under standard lighting conditions. By using contour detection and centroid calculation, the system determines the exact position of objects within the workspace, which is essential for guiding the robotic arm precisely [6].

The coordinate mapping process, implemented using homography transformation, effectively converts pixel coordinates into real-world coordinates. This allows the robotic arm to reach the exact location of the detected object with high precision. As a result, the system performs pick-and-place operations without significant positional errors [8].

The robotic control module, implemented using the Dobot Magician, executes tasks smoothly and efficiently. The robotic arm responds accurately to commands and maintains stable performance during continuous operation. This demonstrates the reliability of the control mechanism in real-time conditions [9].

The integration of all modules ensures proper communication between speech input, vision processing, and robotic execution. The system operates continuously and maintains consistent performance while responding to user commands in real time [4]. Although minor limitations such as sensitivity to extreme lighting conditions and background noise were observed, they do not significantly affect the overall functionality of the system.

Overall, the experimental results confirm that the proposed system achieves good accuracy, reliability, and efficiency. It successfully meets the objective of providing a user-friendly and real-time robotic control system, and it can be effectively applied in areas such as automation, education, and intelligent robotic applications [10].

Voice Command Recognition Output

```

225
226     cv2.imshow("Robot Vision", display)
227
228     if cv2.waitKey(1) & 0xFF == ord('q'):
229         break
230
231
232     cap.release()
233     cv2.destroyAllWindows()
234     device.close()

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

You said: green green
Picking GREEN

Say: Pick Red / Pick Green / Pick Blue

Say: Pick Red / Pick Green / Pick Blue

Say: Pick Red / Pick Green / Pick Blue

Fig 5.4.1 Voice Command Recognition Output

Real-Time Object Detection

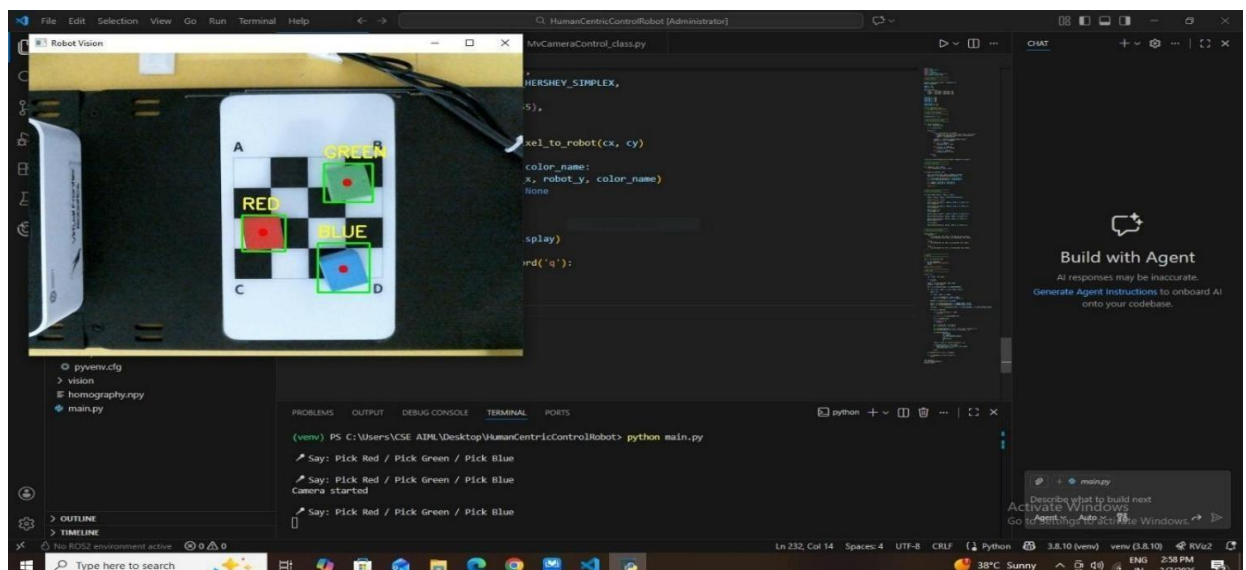


Fig 5.4.2 Real-Time Object Detection

Bounding Box and Centroid Detection

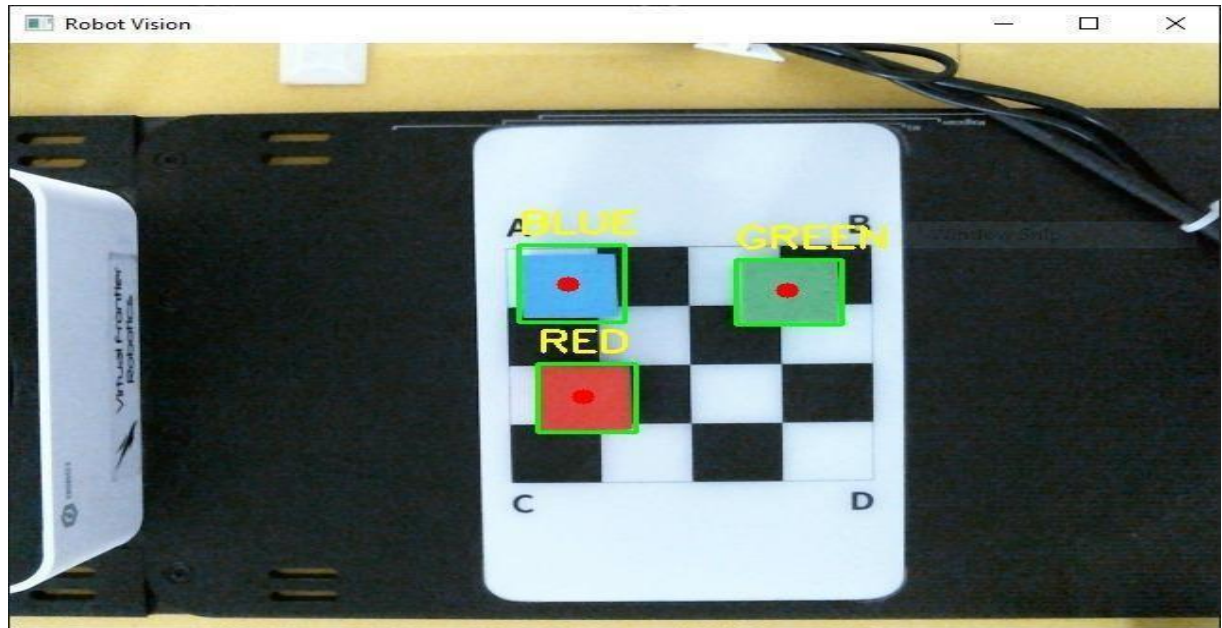


Fig 5.4.3 Bounding Box and Centroid Detection

Robotic Arm Pick and Place Operation



Fig 5.4.4 Robotic Arm Pick and Place Operation

Robotic Arm Pick and Place Operation



Fig 5.4.5 Robotic Arm Pick and Place Operation

Console Output Showing Processing Steps



Fig 5.4.6 Console Output Showing Processing Steps

CHAPTER- 6

CONCLUSION

The proposed human-centric robotic control system effectively combines speech recognition and vision-based object detection to enable a simple and intuitive interaction between humans and robots [10]. The system allows users to operate a robotic arm using voice commands, removing the need for complex manual controls and making the system more accessible and user-friendly [5]. By integrating voice input with visual verification, the system improves both accuracy and reliability during task execution.

The speech recognition module successfully converts spoken commands into text and identifies user intentions with good accuracy under normal conditions. At the same time, the vision module, implemented using OpenCV and HSV-based color segmentation, is capable of detecting objects and determining their positions within the workspace [6]. This combination ensures that commands are executed only after confirming the presence of the required object.

The use of homography mapping plays a crucial role in converting image-based coordinates into real-world coordinates. This allows the robotic arm to move precisely to the target location, improving the accuracy of pick-and-place operations [8]. The robotic control module, implemented using the Dobot Magician, performs tasks smoothly and efficiently, demonstrating reliable real-time performance [9].

All modules in the system work together in a coordinated manner, ensuring smooth communication between input, processing, and execution stages. Although minor limitations such as sensitivity to lighting conditions and background noise were observed, the overall system performance remains stable and effective.

In conclusion, the project provides a practical, efficient, and user-friendly solution for robotic control using natural voice interaction combined with visual validation. The system has strong potential for applications in automation, education, and intelligent robotic systems, and contributes to the development of advanced human–robot interaction technologies [10].

CHAPTER – 7

FUTURE ENHANCEMENTS

The human-based robotic control system provided can also be optimized in different aspects to mediate its work, accuracy and usage in a real world situation.

One of the most significant spheres that should be enhanced is the advanced speech recognition with the assistance of artificial intelligence. Machine learning and deep learning techniques can be added to the system and this will enable it to be more precise in recognizing voice commands. This will help the system identify the different accents, differences in the speech and pronunciation styles more effectively and improve the overall interaction of the user [5].

Another notable enhancement is multi-language voice support. By enabling the system to recognize and handle different languages, it will be easier to make the system more acceptable to the users of other regions. This alternative will facilitate its usage and be more relevant across diverse environments [1].

To enhance the system further, deep learning–based object detection techniques can be integrated. Methods such as Convolutional Neural Networks (CNNs) and YOLO (You Only Look Once) enable the detection of a wider variety of objects with improved accuracy and robustness compared to conventional color-based techniques. This advancement would allow the system to perform more effectively in complex and dynamic environments [6].

The system can also be improved to perform effectively under varying conditions. By making it more robust to changes in lighting, background variations, and noise, its reliability in real-world scenarios can be significantly increased [7].

Another possible enhancement is the use of advanced hybrid intelligence, where speech and vision are combined more effectively. This would allow the system to handle more complex commands by considering both voice input and visual information, such as identifying objects based on their color and position. Such an approach would increase flexibility and enable more intelligent robotic operations [3].

The integration of wireless communication and Internet of Things (IoT) technologies can further extend the system’s capabilities. This would allow users to monitor and control the robotic system remotely through mobile applications or cloud-based platforms, improving accessibility and convenience [4].

Additionally, the system can be adapted for use in industrial automation, where it can assist in performing repetitive tasks efficiently and accurately.

It can be used in performing tasks such as sorting, packaging and assembly tasks therefore can be

used in a real life industrial application. This will bring utility to the system [9].

Finally, user experience can be improved with the implementation of a feedback system that allows the provision of a real-time system. The system can provide audio or visual feedback to inform the users that a task is done, an error has been made or inform about system status. This will enhance communication and the system will be user friendly [10].

All these enhancements will make the system more efficient, intelligent and flexible and can be used more in the future in more advanced robotic applications.

CHAPTER – 8

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